22.4 Mechanical properties of magnesium and magnesium alloys

Related designations for magnesium alloys are given in Chapter 1.

Table 22.18 MAGNESIUM AND MAGNESIUM ALLOYS (WROUGH)—TYPICAL MECHANICAL PROPERTIES AT ROOM TEMPERATURE

			Tension			*Compression*	
Material	*Nominal* composition* %	*Form*	*Proof stress* 0.2% MPa	*UTS* MPa	*Elong.* %	*Proof stress* 0.2% MPa	*Hardness VPN* 30 kg
Mg	Mg 99.9	Sheet, annealed	69	185	4	—	30–35
		Bar, extruded	100	232	6	—	35–45
Mg–Mn	Mn 1.5	Sheet	100	232	6	—	35–45
		Extruded bar (1 in. diam.)	162	263	7	124	45–55
		Extruded tube	154	247	6	—	45–55
Mg–Al–Zn	Al 3.0 Zn 1.0 Mn 0.3	Sheet, annealed	131	232	13	—	50–60
		half hard	170	263	10	100	55–70
		Extruded bar and sections	162	255	11	93	50–60
	Al 6.0 Zn 1.0	Forgings	183	293	8	147	60–70
		Extruded bar and sections	183	293	8	147	55–70
	Mn 0.3	Extruded tube	170	278	8	147	60–70
	Al 8.0 Zn 0.5 Mn 0.3	Forgings	208	293	8	185	65–75
Mg–Zn–Mn	Zn 2.0 Mn 1.0	Sheet, annealed	131	232	13		
		half breed	170	263	10		
		Extruded bar section	162	255	11		
Mg–Zn–Zr	Zn 1.0 Zr 0.6	Sheet	178	263	10	154	55–70
		Extruded bar and sections	208	293	13	177	60–75
		Extruded tube	193	278	7	—	60–75
	Zn 3.0 Zr 0.6	Sheet	185	270	8	154	60–70
		Forgings	224	309	8	193	60–80
		Extruded bar and sections (1 in. diam.)	239	309	18	213	65–75
	Zn 5.5 Zr 0.6	Bars and sections Heat treated	270	340	10	255	60–80
Mg–Zn–Cu–Mn	Zn 6.5 Cu 1.3 Mn 0.8	Bars and sections Heat treated	340	360	6	—	—
Mg–Th–Zn–Zr**	Th 0.8	Extruded bar and sections 5111	147	263	18	—	50–70
(Creep resistant)	Zn 0.5 Zr 0.6	Forgings 5111	147	232	13	—	50–70
Mg–Th–Mn** (Creep resistant)	Th 2.0 Mn 0.75	Sheet	165	247	9	179	—
	Th 3.0 Mn 1.2	Extruded bar and sections	227	287	8	185	—

* It is usual to add 0.2–0.4% Mn to alloys containing aluminium to improve corrosion resistance.
** Thorium-containing alloys are being replaced by alternative Mg alloys.

Table 22.19 MAGNESIUM AND MAGNESIUM ALLOYS (CAST) TYPICAL MECHANICAL PROPERTIES AT ROOM TEMPERATURE

				Tension			*Compression*	
Material	*Nominal* composition* %		*Condition*	*Proof stress* 0.2% MPa	*UTS* MPa	*Elong.* %	*Proof stress* 0.2% MPa	*Brinell hardness*† *VPN* 30 kg
Mg–Zr	Zr	0.6	AC	51	185	2.0	54	40–50
Mg–Al–Zn	Al	6.0	AC	97	199	5	97	50
	Zn	3.0	TB	97	275	10	97	55
			TF	131	275	5	131	73
	Al	8.0	AC	86	158	4	86	50–60
	Zn	0.4	TB	82	247	11	82	50–60
	Al	9.5	AC	93	154	2	93	55–65
	Zn	0.4	TB	90	232	6	90	55–65
			TF	127	239	2	124	75–85
			Die cast	111	216	3	108	60–70
	Al	9.0	AC	97	165	2	97	65
	Zn	2.0	TB	97	275	8	97	63
			TF	145	275	2	145	84
Mg–Zn–Zr	Zn	4.5	TE	161	263	6	162	65–75
	Zr	0.7						
Mg–Zn–RE–Zr	Zn	4.0	TE	150	216	5	139	55–75
	RE	1.2						
	Zr	0.7						
	Zn	6.0	TF§	190	295	7	190	70–80
	RE	2.5						
	Zr	0.7						
Mg–RE–Zn–Zr (Creep resistant to 250°C)	RE	2.7	TE	95	162	4.5	93	50–60
	Zn	2.2						
	Zr	0.7						
Mg–Th–Zn–Zr** (Creep resistant to 350°C)	Th	3.0	TE	93	216	7	93	50–60
	Zn	2.2						
	Z	r 0.7						
Mg–Zn–Th–Zr**	Zn	5.5	TE	167	270	8	162	65–75
	Th	1.8						
	Zr	0.7						
Mg–Th–Zr**	Th	3.0	TF	93	208	5	93	50–60
	Zr	0.7						
Mg–Ag–RE‡–Zr	Ag	2.5	TF	187	247	5	178	65–80
	RE	2.0‡		204	260	3	193	65–80
	Zr	0.6						
	Ag	2.5	TF	200	260	4	195	65–80
	RE	2.0‡						
	Zr	0.6						
Mg–RE(D)–Ag–Zr–Cu	RE(D)	2.2	TF	195	261	4	—	75–90
	Ag	1.5						
	Zr	0.6						
	Cu	0.07						
Mg–Ag–Th–RE‡–Zr**	Ag	2.5	TF	210	270	4	200	65–80
	RE	1.0‡						
	Th	1.0						
Mg–Y–RE(Δ)–Zr	Y	4.0	TF	185	265	7	—	75–90
	Zr	0.7						
	RE(Δ)	3.4						
	Zr	0.6						
	Y	5.1	TF	205	280	4	—	75–90
	RE(Δ)	3.0						
	Zr	0.6						
Mg–Zn–Cu–Mn	Zn	6.0	TF	158	242	4.5	—	55–65
	Cu	2.7						
	Mn	0.5						

* It is usual to add 0.2–0.4% Mn to alloys containing aluminium to improve corrosion resistance. RE = Cerium mischmetal containing approx. 50% cerium. RE(Δ) = Neodymium plus Heavy Rare Earth metals.
RE(D) = Neodymium enriched mischmetal.

† Brinell tests with 500 kg on 10 mm ball for 30 s.

‡ Fractionated rare earth metals: MSR-A contains 1.7%; MSR-B contains 2.5%.

§ Solution heat treated in an atmosphere of hydrogen.

AC = Sand cast. TE = Precipitation heat treated.
TB = Solution heat treated. TF = Fully heat treated.

** Thorium-containing alloys are being replaced by alternative Mg alloys.

Table 22.20 MAGNESIUM AND MAGNESIUM ALLOYS (EXCLUDING HIGH TEMPERATURE ALLOYS FOR WHICH SEE TABLE 22.21)—TYPICAL TENSILE PROPERTIES AT ELEVATED TEMPERATURES

				'Short-time' tension†			
Material	*Nominal composition** %	*Form and condition*	*Test temp.* °C	*Young's modulus* GPa	0.2% *proof stress* MPa	*UTS* MPa	*Elong.* %
Mg	Mg 99.95	Forged	20	45	—	170	5
			100	—	—	128	8
			150	—	—	93	16
			200	—	—	54	43
Mg–Al–Zn	Al 8.0	Sand cast	20	45	86	158	4
	Zn 0.4		100	34	76	154	5
	(A8)		150	32	65	145	11
			200	25	62	100	20
			250	—	—	75	27
		Sand cast	20	45	82	247	11
		and	100	34	73	202	16
		solution	150	33	65	154	21
		treated	200	28	62	116	25
			250	—	—	85	21
	(AZ855)	Forged	20	45	221	309	8
			150	—	153	216	25
			200	—	102	154	28
	Al 9.5	Sand cast	20	45	93	154	2
	Zn 0.4		100	—	—	131	2
	(AZ91)		150	—	—	122	6
			200	—	—	108	25
			250	—	—	77	34
		Sand cast	20	45	90	232	6
		and	100	—	—	222	12
		solution	150	—	—	196	16
		treated	200	—	—	139	20
		Sand cast	20	45	127	239	2
		and	100	40	91	232	6
		fully heat	150	37	77	185	25
		treated	200	28	62	133	34
			250	19	46	103	30
Mg–Zn–Zr	Zn 4.5	Sand cast	20	45	161	263	6
	Zr 0.7	and heat	100	34	124	185	14
	(Z5Z)	treated	150	28	102	145	20
			200	22	79	113	23
			250	19	57	85	20
	Zn 3.0	Extruded	20	45	255	309	18
	Zr 0.6		100	40	162	182	33
	(ZW3)		200	22	46	127	56
			250	12	11	100	71
		Sheet	20	45	195	270	10
			100	40	120	165	33
			150	33	74	116	42
			200	—	—	76	51
			250	—	—	49	59
Mg–Zn–RE–Zr	Zn 4.0	Sand cast	20	45	150	216	4
	Re 1.2	Sand cast	20	41	134	195	6
	Zr 0.7	treated	150	40	120	167	19
	(RZ5)		200	38	99	131	29
			250	33	74	99	35

(*continued*)

Table 22.20 MAGNESIUM AND MAGNESIUM ALLOYS (EXCLUDING HIGH TEMPERATURE ALLOYS FOR WHICH SEE TABLE 22.21)—TYPICAL TENSILE PROPERTIES AT ELEVATED TEMPERATURES—*continued*

				'Short-time' tension†			
Material	*Nominal composition** %	*Form and condition*	*Test temp.* °C	*Young's modulus* GPa	0.2% *proof stress* MPa	*UTS* MPa	*Elong.* %
Mg–Zn–Th–Zr**	Zn 5.5 Th 1.8 Zr 0.7 (TZ6)	Sand cast and heat treated	20 100 150 200 250	45 34 31 28 26	161 134 110 82 52	270 224 178 130 91	9 22 26 26 25
Mg–Ag–RE–Zr (D)	Ag 2.5 RE(D)2.0 Zr 0.6 (QE22)	Sand cast and fully heat treated	20 100 150 200 250 300	45 41 40 38 34 31	201 185 171 154 102 68	259 232 210 185 142 88	4 12 16 20 27 59
Mg–RE(D)–Ag–Zr–Cu	RE(D) 2.2 Ag 1.5 Zr 0.6 Cu 0.07 (EQ21)	Sand cast and fully heat treated	20 100 150 200 250 300	45 43 42 41 39 35	195 189 180 170 152 117	261 230 211 191 169 132	4 10 16 16 15 10
Mg–Ag–RE(D)** Th–Zr‡	Ag 2.6 RE(D) 1.0 Th 1.0 Zr 0.6 (OH21)	Sand cast and fully heat treated	20 100 150 200 250 300	45 41 40 38 37 33	210 199 190 183 167 120	270 242 224 205 185 131	4 17 20 18 19 20
Mg–Y–RE(Δ)–Zr	Y 4.0 RE(Δ) 3.4 Zr 0.6 (WE43)	Sand cast and fully heat treated	20 150 200 250 300	45 42 39 37 35	185 175 170 160 120	265 250 245 220 160	7 6 11 18 40
	Y 5.1 RE(Δ) 3.0 Zr 0.6 (WE54)	Sand cast and fully heat treated	20 100 150 200 250 300	45 43 42 41 39 36	205 197 195 183 175 117	280 260 255 241 230 184	4 4.5 5 6.5 9 14.5
Mg–Zn–Cu–Mn	Zn 6.0 Cu 2.7 Mn 0.5 (Zc63)	Sand cast and fully heat treated	20 100 150 200	45 — — —	158 141 134 118	242 215 179 142	4.5 9 14 11
	Zn 6.5 Cu 1.3 Mn 0.8 (Zc71)	Extruded and fully heat treated	20 100 200	45 40 32	325 206 115	350 259 163	6 16 14

* It is usual to add 0.2–0.4% Mn to alloys containing aluminium to improve corrosion resistance.

† In accordance with BS1094: 1943; 1 h at temperature and strain rate 0.1–0.25 in in^{-1} min^{-1}.

‡ Tested according to BS4A4. RE = Cerium mischmetal containing approx. 50% Ce. RE(D) = Neodymium enriched mischmetal. RE(Δ) = Neodymium plus Heavy Rare Earth metals.

** Thorium-containing alloys are being replaced by alternative Mg alloys.

Table 22.21 HIGH TEMPERATURE MAGNESIUM ALLOYS—TENSILE PROPERTIES AT ELEVATED TEMPERATURE

Material	*Nominal composition** %	*Form and condition*	*Test temp.* °C	*'Short-time' tension*† *Young's modulus* GPa	0.2% *proof stress* MPa	*UTS* MPa	*Elong.* %
Mg–RE–Zn	RE 2.7 Zn 2.2 Zr 0.7 (ZRE1)	Sand cast and heat treated	20	45	93	162	4.5
			100	40	79	150	11
			150	38	76	139	19
			200	36	74	125	26
			250	33	65	107	35
			300	28	48	85	51
			350	21	26	56	90
Mg–Th–Zr**	Th 3.0 Zr 0.7 (HK31) (MTZ)	Sand cast and fully heat treated	20	45	93	208	4
			100	40	88	188	10
			150	38	86	174	13
			200	38	85	162	17
			250	36	83	150	20
			300	34	73	136	22
			350	29	56	103	23
Mg–Th–Zn–Zr**	Th 3.0 Zn 2.2 Zr 0.7 (ZT1)	Sand cast and heat treated	20	45	93	216	9
			100	36	88	159	23
			150	34	79	131	27
			200	33	65	108	33
			250	33	56	90	38
			300	31	49	76	41
			350	28	45	63	34
	Th 0.8 Zn 0.5 Zr 0.6 (ZTY)	Sheet	20	45	181	266	10
			100	41	179	224	10
			150	41	176	201	11
			200	40	165	171	15
			250	40	124	134	20
			300	34	73	96	27
			350	29	17	56	38
Mg–Ag–RE(D)–Zr	Ag 2.5 RE(D) 2.0 Zr 0.6 (QE22)	Sand cast and fully heat treated					
Mg–RE(D)–Ag–Zr–Cu	RE(D) 2.2 Ag 1.5 Zr 0.6 Cu 0.07 (EQ21)	Sand cast and fully heat treated					
Mg–Ag–RE(D)–Th–Zr**	Ag 2.5 RE(D) 1.0 Th 1.0 Zr 0.6 (QH21)	Sand cast and fully heat treated		High strength cast alloys with good elevated temperature properties—for which see Table 22.20			
Mg–Y–RE(Δ)–Zr	Y 4.0 RE(Δ) 3.4 (WE43)	Sand cast and fully treated					
	Y 5.1 RE(Δ) 3.0 Zr 0.6 (WE54)	Sand cast and fully heat treated					

* It is usual to add 0.2–0.4% Mn to alloys containing aluminium to improve corrosion resistance.
† In accordance with BS 1094: 1943; 1 h at temperature; strain rate 0.1–0.25 in in^{-1} min^{-1}.
RE = Cerium mischmetal containing approx. 50% Ce. RE(D) = neodymium-enriched mischmetal.
RE(Δ) = Neodymium plus Heavy Rare Earths.
** Thorium containing alloys are being replaced by alternative Mg alloys.

Table 22.22 HIGH-TEMPERATURE MAGNESIUM ALLOYS—LONG-TERM CREEP RESISTANCE

Material	*Nominal composition %*	*Form and Condition*	*Temp. °C*	*Time*† *h*	*Stress to produce specified creep strains %* 0.05 MPA	0.1 MPa	0.2 MPa	0.5 MPa	1.0 MPa
Mg–RE–Zn–Zr	RE 2.7 Zn 2.2 Zr 0.7 (ZRE1)	Sand cast and heat treated	200	100	52	66	71	—	—
				500	41	54	65	—	—
				1 000	36	47	58	—	—
			250	100	23	28	32	36	—
				500	11	19	24	30	34
				1 000	—	14	20	26	30
			315	100	5.6	7.4	8	—	—
				500	—	5.2	6.5	—	—
				1 000	—	4.3	5.6	—	—
	Zn 4.0 RE 1.2 Zr 0.7 (RZ5)	Sand cast and heat treated	100	100	—	97	111	117	—
				500	—	—	106	117	—
				1 000	—	—	103	116	—
			150	100	77	86	97	101	107
				500	—	75	88	96	100
				1 000	—	70	83	91	97
			200	100	29	43	52	67	73
				500	22	28	37	52	64
				1 000	20	23	31	43	53
			250	100	6.2	12	19	32	39
				500	4.3	6.2	8.6	15	19
				1 000	3.9	5.4	6.9	12	15
Mg–Th–Zr**	Th 3.0 Zr 0.7 (HK31) (MTZ)	Sand cast and fully heat treated	200	100	31*	45*	63*	97*	111*
				1 100	—	—	62*	100*	—
			260	100	—	28*	43*	65*	—
				1 000	—	—	29*	45*	—
			315	100	9.3*	14*	19*	27*	32*
Mg–Th–Zn–Zr**	Th 0.8 Zn 0.5 Zr 0.6 (ZTY)	Sheet	250	100	Stress of 46 MPa (3 tonf in^{-2}) produced 0.03% creep strain Stress of 46 MPa (3 tonf in^{-2}) produced 0.03% creep strain				
Mg–Th–Zn–Zr*	Th 3.0 Zn 2.2 Zr 0.7 (ZT1)	Sand cast and heat treated	250	100	42	50	56	63	66
				500	35	43	51	58	63
				1 000	31	39	48	56	61
			300	100	23	28	35	46	52
				500	19	21	25	36	41
				1 000	17	19	21	32	36
			325	100	14	19	24	29	36
				500	12	13	16	21	25
				1 000	10	12	13	15	20
			350	100	10	12	18	21	23
				500	—	9	10	12	14
				1 000	—	8	8	9	10
			375	100	—	8	11	12	13
				500	—	—	—	8	9
				1 000	—	—	—	—	8
	Zn 5.5 Th 1.8 Zr 0.7 (TZ6)	Sand cast and heat treated	150	100	51	66	82	96	102
				500	36	56	69	85	94
				1 000	26	51	63	80	90
			200	100	26	32	45	56	62
				500	15	22	26	40	49
				1 000	11	17	20	31	40

(*continued*)

Table 22.22 HIGH-TEMPERATURE MAGNESIUM ALLOYS—LONG-TERM CREEP RESISTANCE—*continued*

Material	*Nominal composition* %	*Form and Condition*	*Temp.* °C	*Time*† h	*Stress to produce specified creep strains %* 0.05 MPA	0.1 MPa	0.2 MPa	0.5 MPa	1.0 MPa
Mg–Ag–RE(D)–Zr	Ag 2.5 RE(D) 2.0 Zr 0.6 (QE22)	Sand cast and fully heat treated	200	100	55	74	88	—	—
				500	—	54	65	82	89
				1 000	—	46	56	73	79
			250	100	18	26	33	—	—
				500	—	15	22	28	31
				1 000	—	10	16	22	26
Mg–RE(D)–Ag–Zr–Cu	RE(D) 2.2 Ag 1.5 Zr 0.6 Cu 0.07 (EQ21)	Sand cast and fully heat treated	200	100	—	78	95	116	—
				500	—	57	71	88	—
				1 000	—	48	62	76	—
			250	100	—	29	36	42	—
				500	—	18	22	30	—
				1 000	—	14	19	24	—
Mg–Ag–RE(D)–Th–Zr**	Ag 2.5 RE(D) 1.0 Th 1.0 Zr 0.6 (QH21)	Sand cast and fully heat treated	250	100	22	32	39	—	—
				500	—	20	26	32	36
				1 000	—	—	21	26	30
Mg–Y–RE(Δ)–Zr	Y 4.0 RE(Δ) 3.4 Zr 0.6 (WE43)	Sand cast and fully heat treated	200	100	—	148	161	173	—
				500	—	—	115	148	—
				1 000	—	—	96	139	—
			250	100	—	44	61	—	—
				500	—	—	46	—	—
				1 000	—	—	39	—	—
	Y 5.1 RE(Δ) 3.0 Zr 0.6 (WE54)	Sand cast and fully heat treated	200	100	—	160	165	—	—
				500	—	120	140	—	—
				1 000	—	120	132	—	—
			250	100	—	47	61	81	—
				500	—	43	40	58	—
				1 000	—	16	32	48	—
Mg–Zn–Cu–Mn	Zn 6.0 Cu 2.7 Mn 0.5 (ZC63)	Sand cast and fully heat treated	150	100	—	94	99	104	—
				500	—	82	92	98	—
				1 000	—	74	89	95	—
			200	100	—	60	63	67	—
				500	—	51	55	61	—
				1 000	—	42	49	55	—

* Total strains.
† 4–6 h heating to test temperature followed by 16 h soaking at test temperature.
RE = Cerium mischmetal containing approx. 50% Ce.
RE(D) = Neodymium-enriched mischmetal.
RE(Δ) = Neodymium plus Heavy Rare Earth metals.
** Thorium-containing alloys are being replaced by alternative Mg alloys.

Table 22.23 HIGH-TEMPERATURE MAGNESIUM ALLOYS—SHORT-TERM CREEP RESISTANCE

Material	*Nominal composition* %	*Form and Condition*	*Temp.* °C	*Time*† s	*Stress to produce specified creep strains %* 0.05 MPA	0.1 MPa	0.2 MPa	0.5 MPa	1.0 MPa	*Stress to fracture* MPa
Mg–RE–Zn–Zr	Re 2.7 Zn 2.2 Zr 0.7 (ZRE1)	Sand cast and heat treated	200	30	—	—	98	118	130	136
				60	—	—	97	117	128	134
				600	—	—	96	116	125	129
			250	30	76	84	92	111	123	130
				60	74	83	91	110	120	129
				600	73	82	89	108	114	125
			315	30	52	59	73	80	85	90
				60	51	58	69	76	83	88
				600	42	49	56	62	68	73
	Zn 4.0 RE 1.2 Zr 0.7 (RZ5)	Sand cast and heat treated	200	30	100	107	116	127	—	136
				60	99	105	114	124	—	134
				600	86	99	103	114	—	125
			250	30	86	90	94	99	—	116
				60	83	88	91	96	—	113
				600	71	76	81	86	—	93
			315	30	62	69	76	79	83	86
				60	59	66	73	76	79	82
				600	48	53	59	64	67	69
Mg–Th–Zr*	Th 3.0 Zr 0.7 (HK31) (MTZ)	Sand cast and fully heat treated	250	30	96	103	119	138	—	145
				60	95	103	118	137	—	145
				600	94	102	117	137	—	144
			315	30	80	88	103	117	—	128
				60	78	86	102	116	—	127
				600	74	82	96	107	—	120
Mg–Th–Zn–Zr*	Th 0.8 Zn 0.5 Zr 0.6 (ZTY)	Sheet	250	30	—	—	110	159	163	165
				60	—	—	95	157	160	162
				600	—	—	—	145	149	151
			350	30	20	32	48	80	93	102
				60	18	28	40	67	82	98
				600	—	15	20	31	42	66
Mg–Th–Zn–Zr*	Th 3.0 Zn 2.2 Zr 0.7 (ZT1)	Sand cast and heat treated	200	30	—	—	—	100	118	125
				60	—	—	—	96	114	123
				600	—	—	—	85	103	114
			250	30	58	65	71	84	102	111
				60	57	64	69	81	99	107
				600	56	63	68	74	86	98
			315	30	55	60	64	73	76	82
				60	53	59	63	72	76	80
				600	50	59	61	71	74	77
	Zn 5.5 Th 1.8 Zr 0.7 (TZ6)	Sand cast and heat treated	200	30	96	113	120	128	137	144
				60	93	109	117	124	133	137
				600	63	90	102	110	114	119
			250	30	70	77	85	96	99	107
				60	65	74	80	90	94	99
				600	56	60	66	74	77	82
			315	30	54	59	64	70	74	76
				60	52	57	62	66	70	73
				600	44	49	53	56	58	59

† 1 h heating to test temperature followed by 1 h soaking at test temperature.
RE = cerium mischmetal containing approx. 50% Ce.
* Thorium-containing alloys are being replaced by alternative Mg alloys.

Table 22.24 MAGNESIUM AND MAGNESIUM ALLOYS—FATIGUE AND IMPACT STRENGTHS

					Fatigue strength† at specified cycles						*Impact strength§ for single blow fracture*		
Material	*Nominal* composition %*	*Condition*	‡ *State*	*Temp. °C*	10^5 MPa	5×10^5 MPa	10^6 MPa	5×10^6 MPa	10^7 MPa	5×10^7 MPa	*Test temp. °C*	*Unnotched J*	*Notched J*
Mg–Mn	Mn 1.5 (AM503)	Extruded	U	20	107	90	88	86	85	83	20	12–14	4–4.5
			N		76	90	54	51	50	48			
Mg–Al–Zn	Al 6.0 Zn 1.0 (AZM)	Extruded	U	20	161	139	133	125	124	120	20	34–43	7–9.5
			N		127	110	103	97	94	91			
	Al 8 Zn 0.4 (A8)	Sand cast	U	20	108	93	90	88	88	86	20	3–5	1.5–2
			N		107	80	73	66	65	63			
		Sand cast and solution treated	U	20	124	102	97	91	90	90	20	18–27	4.5–7
			N		108	86	82	74	73	69			
			U	150	93	69	66	59	57	57			
			U	200	71	52	48	38	36	31	−196	1.5	
	Al 9.5 Zn 0.4 (AZ91)	Sand cast	U	20	114	91	89	88	86	85	20	1.5–2.0	1–1.5
			N		110	83	74	68	66	63			
		Sand cast and solution treated	U	20	124	93	93	93	93	—	20	7–9.5	3–4
			N		103	82	80	79	79	77			
		Sand cast and fully heat treated	U	20	117	90	80	79	77	76	20	3–4	1–1.5
			N		93	66	66	65	65	65			
Mg–Zn–Zr	Zn 3.0 Zr 0.6 (ZW3)	Extruded	U	20	151	137	134	128	127	124	20	23–31	9.5–12
			N		124	99	93	91	90	88			
	Zn 4.5 Zr 0.7 (Z5Z)	Sand cast and heat treated	U	20	111	86	85	82	80	77	20	7–12	3–4
			N		90	86	85	82	80	77	−196	0.8	
Mg–Zn–RE–Zr	Zn 4.0 RE 1.2 Zr 0.7 (RZ5)	Sand cast and heat treated	U	20	124	99	97	97	96	94	20	4–5.5	1–2
			N		108	93	91	88	86	83	−196	0.7	
			U	150	97	85	80	73	69	65			
			U	200	93	74	69	62	59	54			

(*continued*)

Table 22.24 MAGNESIUM AND MAGNESIUM ALLOYS—FATIGUE AND IMPACT STRENGTHS—*continued*

			Fatigue strength† at specified cycles							*Impact strength§ for single blow fracture*		
Material	*Nominal* composition* %	*Condition*	‡ *State*	*Temp.* °C	10^5 MPa	5×10^5 MPa	10^6 MPa	5×10^6 MPa	10^7 MPa	5×10^7 MPa	*Test temp.* °C	*Unnotched* J	*Notched* J
	Zn 2.2 RE 2.7 Zr 0.7 (ZRE1)	Sand cast and heat treated	U	20	100	82	80	79	77	74	20	6–7.5	1–2
			N		77	59	54	52	52	51			
			U	150	69	60	59	57	57	57			
			U	200	68	59	56	52	51	51			
			U	250	59	48	45	43	43	42	−196	0.5	
			U	300	49	39	37	37	36	34			
	Zn 6 RE 2.5 Zr 0.6 (ZE63)	Sand cast and fully heat treated**	U	20	144	131	127	121	119	117	20	12.9–17.6	2.3–2.7
			N		99	83	79	73	72	71			
Mg–Ag–RE(D)–Zr	Ag 2.5 RE(D) 2.5 Zr 0.6 (MSR–B)	Sand cast and fully heat treated	U	20	119	103	103	103	102	100			
			N		77	65	63	62	62	62			
			N	200	—	—	—	90	88	86			
			U	250	—	77	68	57	54	51			
Mg–Ag–RE(D)Th–Zr**	Ag 2.5 Re(D) 1.0 Th 1.0 Zr 0.6 (QH21)	Sand cast and fully heat treated	U	20	135	114	111	109	108	108			
			N		86	72	69	64	63	62			
			U	250	108	76	65	56	55	52			
Mg–Zn–Th–Zr**	Zn 5.5 Th 1.8 Zr 0.7 (TZ6)	Sand cast and heat treated	U	20	120	86	85	83	83	82	20	8–11	1.5–3
			N		100	86	80	77	76	76	−196	0.5	
Mg–Th–Zr**	Th 3.0 Zr 0.7 (MTZ)	Sand cast and fully heat treated	U	20	—	74	68	65	63	62			
			N		—	48	40	36	34	32			
			U	200	—	74	68	60	59	58			
			U	250	80	63	59	54	52	51			

Mg–Th–Zn–Zr***	Th	0.7	Extruded	U	20	100	86	83	79	76	74			
	Zn	0.5		N		73	52	51	49	48	46			
	Zr	0.6		U	200	—	74	68	60	59	57			
	(ZTY)			U	250	80	63	59	54	52	51			
	Th	3.0	Sand cast and	U	20	97	82	79	74	71	68	20	7–8	1.5–3
	Zn	2.2	heat treated	N		76	59	56	51	49	48			
	Zr	0.7		U	200	71	60	59	54	52	51			
	(ZT1)			U	250	66	51	46	43	42	39			
				U	325	—	—	43	37	34	29	−196	0.8	
Mg–RE(D)–As–Zr–Cu	RE(D)	2.2	Sand cast and	U	20	103	94	93	92	91	90			
	Ag	1.5	fully heat											
	Zr	0.6	treated											
	Cu	0.07												
Mg–Y–RE(Δ)–YZr	Y	4.0	Sand cast and	U	20	114	101	98	94	93	91			
	RE(Δ)	3.4	fully heat	U	150	107	97	94	87	85	83			
	Zr	0.6	treated	U	250	107	81	74	65	64	62			
	Y	5.9	Sand cast and	U	20	113	104	102	100	99	97			
	RE(Δ)	3.0	fully heat	U	200	118	96	90	84	83	82			
	Zr	0.6	treated	U	250	115	84	78	67	66	65			
Mg–Zn–Cu–Mn	Zn	6.0	Sand cast and	U	20	—	—	100	94	92	90			
	Cu	2.7	fully heat	N		—	—	62	57	56	55			
	Mn	0.5	treated											

* It is usual to add 0.2–0.4% Mn to alloys containing aluminium to improve corrosion resistance.
** Solution heat treated in an atmosphere of hydrogen.
† Wohler rotating beam tests at 2 960 c.p.m.
‡ U = Unnotched.
N = Notched. Semi-circular notch of 0.12 cm (0.047 in) radius. Stress concentration factor 1.8.
§ Hounsfield balanced impact test, notched bar values are equivalent to Izod values.
RE(D) = Neodymium enriched mischmetal.
*** Thorium-containing alloys are being replaced by alternative Mg alloys.
RE(Δ) = Neodymium plus Heavy Rare Earths.

Table 22.25 HEAT TREATMENT OF MAGNESIUM ALLOY CASTINGS

Heat treatment conditions for magnesium sand castings can be varied depending on the particular components and specific properties required. The following are examples of the conditions typically used for each alloy.

Material		*Nominal* composition* %		*Condition*	*Time* h	*Temperature* °C
Mg–Al–Zn	(AZ80)	Al	8.0	TB	12–24	400–420
		Zn	0.4			
	(AZ91)	Al	9.5	TB	16–24	400–420
		Zn	0.4			
	(AZ91)			TF	16–24	400–420
					Air cool	
					8–16	180–210
Mg–Zn–Zr	(Z5Z)	Zn	4.5	TE	10–20	170–200
		Zr	0.7			
Mg–Zn–RE–Zr	(RZ5)	Zn	4.0	TE	2–4	320–340
		RE	1.2		Air cool	
		Zr	0.7		10–20	170–200
	(ZRE1)	RE	2.7	TE	10–20	170–200
		Zn	2.2			
		Zr	0.7			
Mg–Th–Zr†	(HK31)	Th	3.0	TF	2–4	560–570
		Zr	0.7		Air cool	
					10–20	195–205
Mg–Zn–Th–Zr†	(TZ6)	Zn	5.5	TE	2–4	320–340
		Th	1.8		Air cool	
		Zr	0.7		10–20	170–200
	(ZT1)	Th	3.0	TE	10–20	310–320
		Zn	2.2			
		Zr	0.7			
Mg–Ag–RE(D)Zr	(QE22)	Ag	2.5	TF	4–12	520–530
		RE(D)	2.0		Water/Oil Quench	
		Zr	0.6		8–16	195–205
Mg–RE(D)–Ag–Zr–Cu	(EQ21)	RE(D)	2.2	TF	4–12	515–525
		Ag	1.5		Water/Oil Quench	
		Zr	0.6		12–16	195–205
		Cu	0.07			
Mg–Ag–RE(D)–Th–Zr	(QH21)†	Ag	2.5	TF	4–12	520–530
		RE(D)	1.0		Water/Oil Quench	
		Th	1.0		12–20	195–205
		Zr	0.6			
Mg–Y–RE(Δ)–Zr	(WE43)	Y	4.0	TF	4–12	520–530
		RE(Δ)	3.4		Water/Oil Quench	
		Zr	0.6		12–20	245–255
	(WE54)	Y	5.1	TF	4.12	520–530
		RE(Δ)	3.0		Water/Oil Quench/Air Cool	
		Zr	0.6		12–20	245–255
Mg–Zn–Cu–Mn	(ZC63)	Zn	6.0	TF	4–12	435–445
		Cu	2.7		Water Quench	
		Mr	0.5		16–24	180–200

* It is usual to add 0.2–0.4% Mn to alloys containing aluminium to improve corrosion resistance.
RE = Cerium mischmetal containing approximately 50% cerium.
RE(D) = Neodymium-enriched mischmetal.
RE(Δ) = Neodymium plus Heavy Rare Earth metals.
TB = Solution heat treated.
TE = Precipitation heat treated.
TF = Fully heat treated.

† Thorium-containing alloys are being replaced by alternative Mg alloys.

Note: Above 350°C, furnace atmospheres must be inhibited to prevent oxidation of magnesium alloys. This can be achieved either by:
(i) adding $\frac{1}{2}$–1% SO_1 gas to the furnace atmosphere; or
(ii) carrying out the heat treatment in an atmosphere of 100% dry CO_2.

Mechanical properties at subnormal temperatures

At temperatures down to −200°C tensile properties have approximately linear temperature coefficients: proof stress and UTS increase by 0.1–0.2% of the RT value per °C fall in temperature, and elongation falls at the same rate: modulus of elasticity rises approximately 19 MPa (2 800 lbf in^{-2}) per °C over the range 0° to −100°C. No brittle-ductile transitions have been found.

Tests at −70°C have suggested that the magnesium-zinc-zirconium alloys show the best retention of ductility and notched impact resistance at this temperature.

22.5 Mechanical properties of nickel and nickel alloys

Table 22.26 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, STANDARD SPECIFICATIONS AND DESIGNATIONS

Note for 8th edition: In the case of the UK, France, and Germany, this table refers to designations and standards that may have been superceded by European (EN) and international (ISO) standards. Consult Chapter 1 for references to current standards.

Alloy	*Nominal composition %*		*France* AFNOR	*Germany* DIN	Werkstoff Nr.	*UK* BS *and* DTD	*USA* ASTM	ASME	AMS	AECMA	UNS
Nickel 200 201	Ni	99.6	—	17740; Ni 99.2 17750–4	2.406 6/2.406 0 2.406 8/2.406 1	3072–76:NA11 3072–76:NA12	B160–163 B725, B730	SB160–163	5533	—	NO2200
Nickel 205	—	99.6 min.	—	—	2.406 1		F1–3, F9	5555	—		NO2205
Monel 400	Cu Fe Mn	30 1.5 1.0	NU 30	17743: Ni Cu 30 Fe 17750 17751 17752 17753 17754	2.436 0 2.436 1	3072–76:NA13	B127 B163–165 B564	SB127 SB163–165 SB564	4544 4574 4675 4730 4731 7233	—	NO4400
Monel 450	Ni Fe Cu	31.0 0.7 Rem.	—	—	—	—	B111, B122 B151, B171 B359, B395 B432, B466-7 B543, B552	SB111, SB117 SB359, SB395 SN466, SB4467 SB543	—	—	C71500
Monel K-500	Cu Al Ti	29 2.8 0.5	—	17743: Ni Cu 30 Al 17752 17754	2.437 5	3072–76:NA 18	—	Boiler code sect. VIII	4676	—	NO5500
Inconel 600	Cr Fe	16 6	NC 15Fe	17742: Ni Cr 15 Fe 17750 17751 17752 17753 17754	2.481 6	3072–76: NA 14	B163 B166–168 B516 B517 B564 B951	SB163 SB166–168 SB564	5540 5580 5665 5687 7232	—	NO6600
Inconel 601	Ni Cr Al Fe	60.5 23.0 1.4 15.1	—	17742 17750 17751 17752	2.485 1	—	—	Boiler code VIII	5715 5870		NO6601

(*continued*)

Table 22.26 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, STANDARD SPECIFICATIONS AND DESIGNATIONS—*continued*

Alloy	*Nominal composition %*		*France* AFNOR	*Germany* DIN	Werkstoff Nr.	*UK* BS *and* DTD	*USA* ASTM	ASME	AMS	AECMA	UNS
Inconel 617	Ni Cr Co Mo Al C	55.7 21.5 12.5 9.0 1.2 0.1	—	—	—	—	—	Boiler codes I and VIII	—	—	NO6617
Inconel 625	Cr Mo Nb	21.5 9.0 3.6	—	17744 17750 17751 17752 17754	2.485 6	3072 3074 3076 (NA21)	B443 B444 B446 B565 B704 B705 B751	SB443 SB444 SB446 SB564	5666 5599 5837 5581	—	NO6625
Inconel 718	Ni Cr Fe Nb Mo Ti Al	52.5 19.0 18.8 5.2 3.1 0.9 0.5	NC 19 Fe Nb	—	2.466 8	—	B637, B670	Boiler codes I and III	—	Pr En 2404 2405, 2407 2408, 2952 2961, 3219	NO7718
Inconel X-750	Cr Fe Ti Al Nb	15 7 2.5 0.6 0.8	NC 15Fe-T	—	2.466 9	HR 505	B637	SA637	5542 5582/3 5598 5667/8/9 5670/1 568/9 5747, 7246	—	NO7750
Inconel MA 754	Ni Cr Fe Y_2O_3	78.0 20.0 1.0 0.6	—	—	—	—	—	—	—	—	NO7754
INCO 330	Ni Fe Cr Si	35.5 44.8 18.5 1.2	—	—	—	—	B511, B512 B535, B536 B546, B710	SB511, SB536 SB710 Boiler codes VIII, IX	5592 5716	—	NO8330

INCO 020	Ni	35.0	—	—	—	—	B464–4	SB462, SB464	—	—	NO08020
	Fe	38.4					B468	SB468			
	Cr	20.0					B472–4	Boiler codes			
	Cu	3.5					B751	III, VIII, IX			
	Mo	2.5									
	Nb	0.6									
INCO G-3	Ni	49.0	—	17744	2.461 9	—	B581–2	SB581–2	—	—	NO6985
	Cr	22.5		17750–17752			B619, B622	SB619, SB622			
	Fe	19.5					B626, B751	SB626			
	Mo	7.0						Boiler codes			
	Cu	2.0						VIII, IX			
INCO C-276	Ni	59.0	—	17744	2.481 9	—	B574–5	SB574–5	—	—	N10276
	Mo	16.0		17750–17752			B619, B622	SB619, SB622			
	Cr	15.5					B626, B751	SB626			
	Fe	5.5						Boiler codes			
	W	4.0						I, III, VIII, IX			
INCO-HX	Ni	48.3	NC 22 Fe D	—	2.466 5	HR 6, HR 204	B435, B572	SB435, SB572	—	Pr EN	NO6002
	Cr	22.0					B619, B622	SB619, SB622		2182–2185	
	Fe	18.5					B626, B751	SB626			
	Mo	9.0						Boiler codes			
	Co	1.5						III, VIII, IX			
	W	0.6									
	C	0.1									
Incoloy 800	Fe	46	25 NC35–20	X10Ni Cr Al Ti	1.487 6	3072–76:NA 15	B163	SB163	5766	—	NO8800
Incoloy 800 HT	Cr	21		3220			B407–409	SB407–409	5871		NO8811
	Ti	0.4					B564	SB564			
	Al	0.4					B514–5				
							B751				
Incoloy 825	Fe	30	WFe32C20DU	17744	2.485 8	3072–76: NA 16	B163	SB163	—	—	NO8825
	Cr	21		17750–2			B423–425	SB423–425			
	Mo	3		17754			B704–5				
	Cu	2					B751				
	Ti	2									
Incoloy DS	Fe	40	—	—	—	3072–76: NA 17	—	—	—	—	—
	Cr	18									
	Si	2									

(continued)

Table 22.26 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, STANDARD SPECIFICATIONS AND DESIGNATIONS—*continued*

Alloy	*Nominal composition %*	*France* AFNOR	*Germany* DIN	Werkstoff Nr.	*UK* *BS* and DTD	*USA* ASTM	ASME	AMS	AECMA	UNS
Ni-Span C-902	Ni 42 Fe Rem. Cr 5.5 Ti 2.5 Al 0.5	—	—	—	3127	—	—	5210 5221 5223 5225	—	NO9902
Hastelloy B–2	Mo 28	—	—	—	—	B333– B335– B619 B622	SB333 SB335 SB619 SB662	—	—	N10665
Hastelloy C–4	Mo 16 Cr 16	—	—	—	—	B574 B575 B619 B662	SB574 SB575 SB619 SB662	—	—	NO6455
Hastelloy X	Mo 9 Cr 21 Fe 18	NC 22FeD	LW2.4665	—	HR6 HR204	B435 B572 B619 B622 B626	—	5536, 5587 5754, 5588	Ni–P93–HT	NO6002
Nimonic 75	Cr 20 Ti 0.4	NC 20T	17742: Ni Cr 20 Ti LW2.4630 17750–2	2.495 1 2.463 0	Hr5 HR203 HR403 HR504	—	—	—	Pr EN 2293–4 2302 2306–8 2402 2411	NO6075
Nimonic 80A	Cr 20 Ti 2.0 Al 1.5	NC 20TA	Ni Cr 20 Ti Al 17742 17754	2.463 1 2.495 2	3076: NA 20 HR1 HR201 HR401 HR601	B637	—	—	Pr EN 2188–91 2396 2397	NO7080
Nimonic 90	Cr 20 Co 17 Ti 2.4 Al 1.4	NCK 20TA	—	2.463 2	HR2 HR202 HR402 HR501 HR502–3 BS3975–NA19	—	—	5829	Pr EN 2295–99 2401–2 2412 2669–70	NO7090

Nimonic 105	Cr Co Mo Al Ti	15 20 5 5 1.2	MK 20CDA	—	2.463 4	HR3	—	—	—	Pr EN 2179–81	—
Nimonic 115	Cr Co Mo Al Ti	14 13 3 5 4	NK 20ATD	—	2.463 6	HR4	—	—	—	Pr EN 2196 2197	—
Nimonic 263	Cr Co Mo Ti Al	20 20 6 2 0.5	NCK 20D	—	2.465 0	HR10 HR206 HR404	—	—	5872	Pr EN 2199–2203 2418	NO7263
Nimonic 901	Fe Cr Mo Ti	35 13 6 3	Z8 NC DT 42	—	2.466 2	HR53	—	—	5660 5661	Pr EN 2176–8	NO9901
Nimonic PE16	Fe Cr Mo Ti Al	32 16 3 1.0 1.0	NW 11 AC	—	—	HR55 HR207	—	—	—	—	—
Nimonic PK33	Cr Co Mo Ti Al	18 14 7 2.25 2.1	NC 19 KDu/v	—	—	DTD5057	—	—	—	—	—
Astroloy	Ni Cr Co Mo Al Ti	54.8 15.0 17.0 5.3 4.0 3.5	—	—	—	—	—	—	—	—	—

(*continued*)

Table 22.26 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, STANDARD SPECIFICATIONS AND DESIGNATIONS—*continued*

Alloy	*Nominal composition %*		*France* AFNOR	*Germany* DIN	Werkstoff Nr.	*UK* BS *and* DTD	*USA* ASTM	ASME	AMS	AECMA	UNS
Rene 41	Ni	55.4	—	—	—	—	—	—	—	—	NO7041
	Cr	19.0									
	Co	11.0									
	Mo	11.0									
	Al	1.5									
	Ti	3.1									
Rene 95	Ni	61.5	—	—	—	—	—	—	—	—	—
	Cr	14.0									
	Co	8.0									
	Mo	3.5									
	Nb	3.5									
	Al	3.5									
	Ti	2.5									
Rene 100	Ni	61.8	—	—	—	—	—	—	—	—	—
	Cr	9.5									
	Co	15.0									
	Mo	3.0									
	Al	5.5									
	Ti	4.2									
	V	1.0									
Udimet 500	Ni	53.7	—	—	—	—	—	—	—	—	NO7500
	Cr	18.0									
	Co	18.5									
	Mo	4.0									
	Al	2.9									
	Ti	2.9									
Udimet 700	Ni	55.5	—	—	—	—	—	—	—	—	—
	Cr	15.0									
	Co	17.0									
	Mo	5.0									
	Al	4.0									
	Ti	3.5									
Waspaloy	Ni	58.7	—	—	—	—	—	—	—	—	NO7001
	Cr	19.5									
	Co	13.5									
	Mo	4.3									
	Al	1.3									
	Ti	3.0									

Table 22.27 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, MECHANICAL PROPERTIES[1] AT ROOM TEMPERATURES

Alloy[3]	*Nominal composition* %		*Condition*	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %	*Brinell hardness*	*Izod impact* J
Nickel–pure	Ni	99.9	Annealed	60	310	40	85	160
Nickel–comc.	Ni	99.0	Annealed	150	400	40	100	160
			Hot rolled	200	500	40	120	160
			Cold drawn	480	660	25	190	—
Nickel 205	Ni	99.6 min.	Annealed	90	345	45	80	—
Monel[1] 400	Cu	31.5	Annealed	240	550	40	125	140
	Fe	1.5	Hot rolled	490	620	35	150	140
	Mn	1.0	Cold drawn	570	700	25	190	110
Monel 450	Ni	31.0	Annealed	165	385	46	90	—
	Fe	0.7						
	Cu	Rem.						
Monel K-500	Cu	29	Annealed	790	1 100	30	290	—
	Al	2.8	and aged					
	Ti	0.5	Hot rolled	880	1 110	25	310	—
			and aged					
			Cold drawn	850	1 110	20	310	—
			and aged					
Cupro-nickel[4]	Cu	55	Annealed	200	400	45	—	—
			Cold drawn	450	590	20	—	—
Inconel[1] 600	Cr	15.5	Annealed	310	655	45	150	160
	Fe	8	Cold drawn	700	880	20	200	110
Inconel 601	Ni	60.5	Annealed	210–340	550–790	70–40	110–150	—
	Cr	23.0						
	Al	1.4						
	Fe	15.1						
Inconel 617	Ni	55.7	Annealed	350	760	58	173	—
	Cr	21.5						
	Co	12.5						
	Mo	9.0						
	Al	1.2						
	C	0.1						
Inconel 625	Cr	22	Annealed	520	930	45	186	—
	Nb	4	Solution	352	810	50	157	—
	Mo	9	treated					
Inconel 718	Ni	52.5	Precipitation	1 180	1 350	17	382	—
	Cr	19.0	hardened					
	Fe	18.8						
	Nb	5.2						
	Mo	3.1						
	Ti	0.9						
	Al	0.5						
Inconel X-750	Cr	15	Fully heat	900	1 240	20	382	—
	Fe	7	treated					
	Ti	2.5						
	Al	0.6						
	Nb	0.8						
Inconel	Ni	78.0						
MA 754	Cr	20.0						
	Fe	1.0						
	Y_2O_3	0.6						
INCO 330	Ni	35.5	Annealed	270	590	47	150	—
	Fe	44.8						
	Cr	18.5						
	Si	1.2						
INCO 020	Ni	35.0	Annealed	310	620	40	183	—
	Fe	38.4						
	Cr	20.0						
	Cu	3.5						
	Mo	2.5						
	Nb	0.6						

(*continued*)

Table 22.27 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, MECHANICAL PROPERTIES[1] AT ROOM TEMPERATURES—*continued*

Alloy[3]	*Nominal composition* %	*Condition*	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %	*Brinell hardness*	*Izod impact* J
INCO G-3	Ni 49.0 Cr 22.5 Fe 19.5 Mo 7.0 Cu 2.0	Annealed	320	690	50	146	—
INCO C-276	Ni 59.0 Mo 16.0 Cr 15.5 Fe 5.5 W 4.0	Annealed	415	790	50	184	—
INCO-HX	Ni 48.3 Cr 22.0 Fe 18.5 Mo 9.0 Co 1.5 W 0.6 C 0.1	Annealed	340	790	45	184	—
Incoloy[1] 800	Fe 46 Cr 21 Ti 0.4 Al 0.4	Annealed Cold drawn	310 700	590 880	45 20	180 250	160 110
Incoloy 825	Fe 32 Cr 21 Mo 3 Cu 2 Ti 1	Annealed	340	650	40	150	160
Incoloy DS	Fe 40 Cr 18 Si 2	Annealed Cold drawn	220 850	600 1 000	61 10	190 300	140 110
Ni-Span[1] C-902	Fe 47 Cr 5.5 Ti 2.5 Al 0.5	Fully heat treated	770	1 200	25	300	—
Hastelloy B-2	Mo 28	Solution treated	526	955	53	235	—
Hastelloy C-4	Mo 16 Cr 16	Solution treated	416	768	52	184	—
Hastelloy C-276	Mo 16 Cr 15 W 4 Fe 5 C 0.02	Solution treated	360	790	60	200	160
Hastelloy X	Mo 9 Cr 21 Fe 18	Solution treated	350	800	45	175	75
Brightvay C	Cr 20 Si 1	Annealed	330	760	40	160	—
Nimonic[1] 75	Cr 20 Ti 0.4	Annealed	240	750	40	170	110
Nimonic 80A	Cr 20 Ti 2.0 Al 1.5	Fully heat treated	780	1 220	30	370	70
Nimonic 90	Cr 20 Co 17 Ti 2.4 Al 1.4	Fully heat treated	750	1 175	30	380	70

(*continued*)

Table 22.27 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, MECHANICAL PROPERTIES[1] AT ROOM TEMPERATURES—*continued*

Alloy[3]	*Nominal composition* %	*Condition*	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %	*Brinell hardness*	*Izod impact* J
Nimonic 105	Cr 15 Co 20 Mo 5 Al 5 Ti 1.2	Fully heat treated	780	1 140	22	380	16
Nimonic 115	Cr 14 Co 13 Mo 3 Al 5 Ti 4	Fully heat treated	865	1 240	27	400	—
Nimonic 263	Cr 20 Co 20 Mo 6 Ti 2 Al 0.5	Fully heat treated	580	970	39	320	—
Nimonic 901	Fe 35 Cr 13 Mo 6 Ti 3	Fully heat treated	900	1 220	15	—	—
Nimonic PE16	Fe 32 Cr 16 Mo 3 Ti 1.0 Al 1.0	Fully heat treated	460	850	30	280	—
Nimonic PK 33	Cr 18 Co 14 Mo 7 Ti 2.2 Al 2.1	Fully heat treated	790	1 170	30	—	—
Astroloy	Ni 54.8 Cr 15.0 Co 17.0 Mo 5.3 Al 4.0 Ti 3.5	Precipitation hardened	1 050	1 415	16	—	—
Rene 41	Ni 55.4 Cr 19.0 Co 11.0 Mo 11.0 Al 1.5 Ti 3.1	Precipitation hardened	1 060	1 420	14	—	—
Rene 95	Ni 61.5 Cr 14.0 Co 8.0 Mo 3.5 Nb 3.5 Al 3.5 Ti 2.5	Precipitation hardened	1 310	1 620	15	—	—
Udimet 500	Ni 53.7 Cr 18.0 Co 18.5 Mo 4.0 Al 2.9 Ti 2.9	Precipitation hardened	840	1 310	32	—	—

(*continued*)

Table 22.27 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, MECHANICAL PROPERTIES[1] AT ROOM TEMPERATURES—*continued*

Alloy[3]	*Nominal composition* %	*Condition*	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %	*Brinell hardness*	*Izod impact* J
Udimet 700	Ni 55.5 Cr 15.0 Co 17.0 Mo 5.0 Al 4.0 Ti 3.5	Precipitation hardened	965	1 410	17	—	—
Waspaloy	Ni 58.5 Cr 19.5 Co 13.5 Mo 4.3 Al 1.3 Ti 3.0	Precipitation hardened	795	1 275	25	—	—

(1) Registered Trade Mark.
(2) All values in the tables in this section are representative and not to be considered as minima or maxima not to be used for specification purposes. Conversion figures have all been rounded off and are not to be taken as precise equivalents.
(3) Where trade marks apply to the name of an alloy there may be materials of similar composition available from other producers who may or may not use the same suffix along with their own trade names. The suffix alone, e.g. Alloy 800 is sometimes used as a descriptive term for the type of alloy but trade marks can be used only by the registered user of the mark.
(4) Other copper–nickel alloys, cupro-nickels, will be found listed under copper base alloys.
(5) Fully heat-treated usually implies a solution treatment followed by some form of precipitation hardening cycle. The precise heat-treatment and the associated properties may be varied to suit particular applications or according to the form of the alloy as forging, bar, sheet or welded fabrication.
(6) There are many nickel chromium alloys with and without iron used for electrical resistance heating and for general high-temperature applications in furnaces and heat-treatment plant. This alloy is typical.

Table 22.28 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, SHORT-TIME HIGH-TEMPERATURE TENSILE PROPERTIES

Alloy	*Nominal composition* %	*Condition*	*Test temperature* °C	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %
Nickel	Ni 99.0	Hot rolled	20	170	490	50
			200	150	540	50
			400	140	540	50
			600	110	250	60
			800	—	170	60
Nickel 205	Ni 99.6 min.	Annealed	20	90	345	45
Monel* 400	Cu 30 Fe 1.5 Mn 1.0	Hot rolled	20	230	560	45
			200	200	540	50
			400	220	460	52
			600	120	260	30
			800	77	120	54
Monel 450	Ni 31.0 Fe 0.7 Cu Rem.	Annealed	20	165	385	46
			200	120	300	55
			400	100	290	42
			600	75	150	40
			800	28	58	19
Monel* K-500	Cu 29 Al 2.8 Ti 0.5	Fully heat treated	20	340	680	45
			200	290	650	40
			400	260	600	30
			600	290	460	5
			800	—	185	30

(*continued*)

Table 22.28 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, SHORT-TIME HIGH-TEMPERATURE TENSILE PROPERTIES—*continued*

Alloy	*Nominal composition* %	*Condition*	*Test temperature* °C	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %
Inconel* 600	Cr 16 Fe 6	Hot rolled	20 400 600 800 1 000	250 185 150 95 —	590 560 530 250 110	50 50 10 20 50
Inconel 601	Ni 60.5 Cr 23.0 Al 1.4 Fe 15.1	Annealed	20 200 400 600 800 870	440 410 380 325 170 100	760 750 740 575 205 160	43 38 38 32 92 122
Inconel 617	Ni 55.7 Cr 21.5 Co 12.5 Mo 9.0 Al 1.2 C 0.1	Solution Annealed	20 200 400 600 800 1 000	330 260 230 220 230 80	720 640 600 580 410 160	63 70 72 58 65 92
Inconel 718	Ni 52.5 Cr 19.0 Fe 18.8 Nb 5.2 Mo 3.1 Ti 0.9 Al 0.5	Precipitation hardened	20 200 400 600 800	1 180 1 100 1 080 960 720	1 350 1 280 1 240 1 160 760	17 16 13 16 8
Inconel X-750	Cr 15 Fe 7 Ti 2.5 Al 0.6 Nb 0.8	Fully heat treated	20 400 650 800 800	620 590 280 310 100	1 110 1 100 830 370 170	24 28 9 22 90
Inconel MA 754	Ni 78.0 Cr 20.0 Fe 1.0 Y_2O_3 0.6	Annealed	20 200 400 600 800 1 000	560 560 540 500 230 120	940 910 860 640 280 160	20 18 16 18 30 16
INCO 330	Ni 35.5 Fe 44.8 Cr 18.5 Si 1.2	Annealed	20 200 400 600 800 980	280 220 200 165 120 70	580 520 500 430 170 80	48 45 44 49 68 74
INCO 020	Ni 35.0 Fe 38.4 Cr 20.0 Cu 3.5 Mo 2.5 Nb 0.6	Annealed	30 200 400 600 800 870	300 280 250 185 150 140	620 575 560 505 275 220	40 40 39 41 59 72
INCO G-3	Ni 49.0 Cr 22.5 Fe 19.5 Mo 7.0 Cu 2.0	Solution Annealed	20 200 400 600 800 1 000	320 235 190 175 170 75	690 590 555 500 355 140	50 62 68 73 66 76

(*continued*)

Table 22.28 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, SHORT-TIME HIGH-TEMPERATURE TENSILE PROPERTIES—*continued*

Alloy	*Nominal composition* %		*Condition*	*Test temperature* °C	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %
INCO C-276	Ni	59.0	Annealed	20	415	790	50
	Mo	16.0		200	285	690	56
	Cr	15.5		400	240	660	64
	Fe	5.5		540	220	610	60
	W	4.0					
INCO-HX	Ni	48.3	Solution	20	340	790	45
	Cr	22.0	Annealed	200	300	745	40
	Fe	18.5		400	225	700	48
	Mo	9.0		600	220	600	46
	Co	1.5		800	180	370	48
	W	0.6		1 000	40	110	56
	C	0.1					
Incoloy* 800	Fe	45	Annealed	20	300	600	45
	Cr	21		600	210	440	40
	Ti	0.4					
	Al	0.4					
Incoloy DS	Fe	40	Annealed	20	300	700	45
	Cr	18		600	—	460	50
	Si	2		800	—	150	90
				1 000	—	75	100
Hastelloy* B-2	Mo	28	Solution treated	200	451	885	50
				320	426	864	49
				430	418	866	51
Hastelloy C-4	Mo	16	Solution treated	200	403	706	49
	Cr	16		320	371	675	52
				430	320	656	64
Hastelloy X	Mo	9	Solution treated	20	360	790	45
	Cr	21		600	280	620	40
	Fe	18		800	230	410	37
				1 000	110	160	45
Nimonic* 75	Cr	20	Annealed	20	420	800	35
	Ti	0.4		400	400	740	30
				600	310	590	30
				800	110	220	80
				1 000	50	90	58
Nimonic* 80A	Cr	20	Fully heat	20	740	1 240	24
	Ti	2.0	treated	400	680	1 150	26
	Al	1.5		600	620	1 080	20
				800	490	620	24
				1 000	70	120	120
Nimonic 90	Cr	20	Fully heat	20	750	1 175	30
	Co	17	treated	600	680	1 030	26
	Ti	2.4		800	530	900	18
	Al	1.4		1 000	48	76	130
Nimonic 105	Cr	15	Fully heat	20	780	1 140	22
	Co	20	treated	600	720	1 040	25
	Mo	5		800	680	810	25
	Al	5		1 000	150	175	42
	Ti	1.2					

(*continued*)

Table 22.28 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, SHORT-TIME HIGH-TEMPERATURE TENSILE PROPERTIES—*continued*

Alloy	*Nominal composition* %		*Condition*	*Test temperature* °C	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %
Nimonic 115	Cr	14	Fully heat	20	860	1 230	27
	Co	13	treated	600	790	1 100	20
	Mo	3		800	760	1 020	19
	Al	5		1 000	200	420	26
	Ti	4					
					0.1% PS		
Nimonic 263	Cr	20	Fully heat	20	570	970	40
	Co	20	treated	600	460	790	41
	Mo	6		800	390	560	20
	Ti	2		1 000	60	110	65
	Al	0.5					
Nimonic PE 16	Cr	16	Fully heat	20	490	880	37
	Fe	32	treated	600	450	730	27
	Mo	3		800	290	390	53
	Ti	1.0		1 000	46	92	100
	Al	1.0					
Astroloy	Ni	54.8	Precipitation	20	1 050	1 410	16
	Cr	15.0	hardened	540	965	1 240	16
	Co	17.0		650	965	1 310	18
	Mo	5.3		760	910	1 160	21
	Al	4.0		870	690	770	25
	Ti	3.5					
Rene 41	Ni	55.4	Precipitation	21	1 060	1 420	14
	Cr	19.0	hardened	540	1 010	1 400	14
	Co	11.0		650	1 000	1 340	14
	Mo	11.0		760	940	1 100	11
	Al	1.5		870	550	620	19
	Ti	3.1					
Rene 95	Ni	61.5	Precipitation	21	1 310	1 620	15
	Cr	14.0	hardened	540	1 250	1 540	12
	Co	8.0		650	1 220	1 460	14
	Mo	3.5		760	1 100	1 170	15
	Nb	3.5					
	Al	3.5					
	Ti	2.5					
Udimet 500	Ni	53.7	Precipitation	21	840	1 310	32
	Cr	18.0	hardened	540	765	1 240	28
	Co	18.5		650	760	1 210	28
	Mo	4.0		760	730	1 040	39
	Al	2.9		870	495	640	20
	Ti	2.9					
Udimet 700	Ni	55.5	Precipitation	21	965	1 410	17
	Cr	15.0	hardened	540	895	1 280	16
	Co	17.0		650	855	1 240	16
	Mo	5.0		760	825	1 030	20
	Al	4.0		870	635	690	27
	Ti	3.5					
Waspaloy	Ni	58.7	Precipitation	21	795	1 280	25
	Cr	19.5	hardened	540	725	1 170	23
	Co	13.5		650	690	1 120	34
	Mo	4.3		760	675	795	28
	Al	1.3		870	515	525	35
	Ti	3.0					

* Registered trade mark.

Table 22.29 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, CRYOGENIC PROPERTIES

Alloy	*Nominal composition* %	*Condition*	*Test temp.* °C	0.2% *Proof stress* MPa	*UTS* MPa	*Elongation* %	*Charpy impact* J	*Notched fatigue strength** MPa
Nickel	Ni 99.0	Annealed	20	160	500	48	230	—
			−75	170	560	58	230	120
			−200	230	710	54	230	140
Monel** 400	Cu 30	Annealed	20	220	540	52	260	—
	Fe 1.5		−180	340	800	50	240	—
	Mn 1.0							
Monel** K-500	Cu 29	Fully heat	20	670	1 060	22	74	—
	Al 2.8	treated	−80	740	1 140	24	68	300
	Ti 0.5		−200	830	1 260	30	—	330
			−255	940	1 380	28	—	330
Inconel** 600	Cr 16	Annealed	20	230	650	37	240	280
	Fe 6		−80	290	730	40	210	280
			−190	—	—	—	190	300
Inconel 718	Cr 19	Fully heat	20	1 400	1 500	—	—	—
	Nb 5	treated	−80	1 400	1 600	—	—	—
	Mo 3		−190	1 600	1 800	—	—	—
	Ti 0.9							
	Al 0.5							
Inconel X-750	Cr 15	Fully heat	20	700	1 200	24	—	—
	Fe 7	treated	−75	790	1 290	23	—	410
	Ti 2.5		−200	810	1 440	19	—	440
	Al 0.6		−255	900	1 450	16	—	460
	Nb 0.8							
Ni-Span** C-902	Fe 47	Fully heat	20	760	1 210	30	24	320
	Cr 5.5	treated	−130	860	1 410	28	23	310
	Ti 2.5		−255	1 000	1 690	25	23	390
	Al 0.5							

* 10^6 cycles Kt = 3.1. ** Registered trade mark.

Table 22.30 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, FATIGUE PROPERTIES

Alloy	*Nominal composition* %	*Condition*	*Test temperature* °C	*Average endurance limit Rotating beam tests* 10^8 cycles MPa
Nickel	Ni 99.0	Annealed	20	230
		Cold drawn	20	340
Monel** 400	Cu 30	Annealed	20	230
	Fe 1.5	Cold drawn*	20	300
	Mn 1.0			
Monel K-500	Cu 29	Annealed	20	260
	Al 2.8	Cold drawn and	20	320
	Ti 0.5	aged		
Inconel** 600	Cr 16	Annealed	20	270
	Fe 6		650	180
			850	75
Inconel 625	Cr 21.5	Hot rolled	29	460
	Mo 9.0		427	450
			538	425
	Nb 3.6		649	390
			760	310

(*continued*)

Table 22.30 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, FATIGUE PROPERTIES—*continued*

Alloy	*Nominal composition* %	*Condition*	*Test temperature* °C	*Average endurance limit Rotating beam tests* 10^8 cycles MPa		
Inconel 718	Cr 19 Nb 5 Mo 3 Ti 0.9 Al 0.5	Fully heat treated	20 650	620 500		
Inconel X-750	Cr 15 Fe 7 Ti 2.5 Al 0.6 Nb 0.8	Fully heat treated	20 700	280 340		
Incoloy** 800	Fe 45 Cr 21 Ti 0.4	Annealed	20 500 850	290 260 95		
				0 ± *stress. For lives of*		
				MPa	h	cycles
Nimonic** 75	Cr 20 Ti 0.4	Annealed	20 750	260 190	65 300	10×10^6 10×10^6
Nimonic 90	Cr 20 Co 17 Ti 2.4 Al 1.4	Fully heat treated	750 870 	260 240 140 110	300 1 000 300 1 000	36×10^6 120×10^6 36×10^6 120×10^6
Nimonic 105	Cr 15 Co 20 Mo 5 Al 5 Ti 1.2	Fully heat treated	20 750 870 980	350 250 260 250 240 190 170 124	50 500 50 500 50 500 50 500	3×10^7 3×10^8 3×10^7 3×10^8 3×10^7 3×10^8 3×10^7 3×10^8

* Stress equalised 3 h at 260°C. **Registered trade marks.

Table 22.31 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, CREEP PROPERTIES

Alloy	*Nominal composition* %	*Condition*	*Test temp.* °C	*Stress* MPa *for creep extension of*								
				0.1%			0.2%			1%	*Rupture*	
				300 h	1 000 h	10 000 h	300 h	1 000 h	10 000 h	10 000 h	1 000 h	1 0000 h
Nickel	Ni 99.0	Cold drawn	400	—	—	77	—	—	—	220	—	—
Monel* 400	Cu 30 Fe 1.5 Mn 1.0	Hot rolled	400	—	—	140	—	—	—	215	—	—
Monel K-500	Cu 29 Al 2.8 Ti 0.5	Cold drawn and aged	400	—	—	355	—	—	—	—	—	—
Inconel* 600	Cr 16 Fe 6	Hot rolled	400 600	—	—	340 38	—	—	—	— 77	—	—
Inconel 601	Ni 60.5 Cr 23.0 Al 1.4 Fe 15.1	Annealed	650 760 870 980 1 095	195 63 30 14 7	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —

(*continued*)

Table 22.31 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, CREEP PROPERTIES—*continued*

Alloy	*Nominal composition* %	*Condition*	*Test temp.* °C	*Stress* MPa *for creep extension of* 0.1% 300 h	0.1% 1 000 h	0.1% 10 000 h	0.2% 300 h	0.2% 1 000 h	0.2% 10 000 h	1% 10 000 h	*Rupture* 1 000 h	*Rupture* 1 0000 h
Inconel 617	Ni 55.7 Cr 21.5 Co 12.5 Mo 9.0 Al 1.2 C 0.1	Solution Annealed	650 760 870 980 1 095	320 150 58 25 10	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —
Inconel 718	Ni 52.5 Cr 19.0 Fe 18.8 Nb 5.2 Mo 3.1 Ti 0.9 Al 0.5	Precipitation hardened	595 650 705 760	760 590 370 170	— — — —	— — — —	— — — —	— — — —	— — — —	— — — —	— — — —	— — — —
Incoloy* 800H	Fe 45 Cr 21 Tl 0.4 Al 0.4	Solution treated	650 870 980	— — —	115 24 7	90 20 3.8	— — —	— — —	— — —	— — —	— — —	121 24 8.3
Inconel X-750	Cr 15 Fe 7 Ti 2.5 Al 0.6 Nb 0.8	Fully heat treated	595 650 730 815 870	— — — — —	— — — — —	— — — — —	— — — — —	— 420 — 110 —	— — — — —	— — — — —	630 470 260 110 50	— — — — —
Inconel MA 754	Ni 78.0 Cr 20.0 Fe 1.0 Y_2O_3 0.6	Annealed	650 760 870 980 1 095 1 150	256 199 158 129 94 78	— — — — — —	— — — — — —	— — — — — —	— — — — — —	— — — — — —	— — — — — —	— — — — — —	— — — — — —
INCO 330	Ni 35.5 Fe 44.8 Cr 18.5 Si 1.2	Annealed	760 870 980 1 095	48 21 8.6 5.4	— — — —	— — — —	— — — —	— — — —	— — — —	— — — —	— — — —	— — — —
INCO-HX	Ni 48.3 Cr 22.0 Fe 18.5 Mo 9.0 Co 1.5 W 0.6 C 0.1	Solution Annealed	760 815 870 925 980	110 72 45 26 15	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —	— — — — —
Nimonic* 75	Cr 20 Ti 0.4	Annealed	650 750 815 870 925 980	56 25 — — — —	— — — — — —	— — — — — —	— — — — — —	62 29 — — — —	51 25 — — — —	— — — — — —	170 50 24 15 10 8	— — — — — —
Nimonic 80A	Cr 20 Ti 2.0 Al 1.5	Fully heat treated	650 705 750 815	390 — 170 93	185 — 120 58	— — 51 23	460 — 200 108	390 — 150 73	260 — 66 26	280 — 90 29	500 350 220 110	280 — 85 34
Nimonic 90	Cr 20 Co 17 Ti 2.4 Al 1.4	Fully heat treated	650 705 750 815 870	400 — 185 85 49	348 — 137 57 29	250 — 65 28 15	450 — 220 108 60	400 — 175 77 37	310 — 85 31 17	325 — 110 38 18	455 360 240 150 75	340 — 135 54 28

(*continued*)

Table 22.31 WROUGHT NICKEL AND HIGH NICKEL ALLOYS, CREEP PROPERTIES—*continued*

Alloy	*Nominal composition* %	*Condition*	*Test temp.* °C	*Stress* MPa *for creep extension of* 0.1% 300 h	0.1% 1 000 h	0.1% 10 000 h	0.2% 300 h	0.2% 1 000 h	0.2% 10 000 h	1% 10 000 h	*Rupture* 1 000 h	*Rupture* 1 0000 h
Nimonic 105	Cr 15 Co 20 Mo 5 Al 5 Ti 1.2	Fully heat treated	750	285	230	—	325	280	193	—	363	208
			815	162	119	—	193	148	93	102	224	120
			870	86	57	—	108	83	34	45	134	57
			940	31	22	—	48	31	12	22	62	35
			980	—	—	—	—	—	—	—	32	—
Nimonic 115	Cr 14 Co 13 Mo 3 Al 5 Ti 4	Fully heat treated	750	355	310	—	385	340	—	—	450	250
			815	—	—	—	—	—	—	—	310	—
			850	154	116	51	222	137	53	74	—	108
			870	—	—	—	—	—	—	—	210	—
			950	51	34	14	63	43	15	17	—	52
			925	—	—	—	—	—	—	—	130	—
			980	—	—	—	—	—	—	—	80	—
			1 000	22	15	—	29	19	6	8	—	—
Astroloy	Ni 54.8 Cr 15.0 Co 17.0 Mo 5.3 Al 4.0 Ti 3.5	Precipitation hardened	650	770	—	—	—	—	—	—	—	—
			760	425	—	—	—	—	—	—	—	—
			870	170	—	—	—	—	—	—	—	—
			980	55	—	—	—	—	—	—	—	—
Rene 41	Ni 55.4 Cr 19.0 Co 11.0 Mo 11.0 Al 1.5 Ti 3.1	Precipitation hardened	650	705	—	—	—	—	—	—	—	—
			760	345	—	—	—	—	—	—	—	—
			870	115	—	—	—	—	—	—	—	—
Rene 95	Ni 61.5 Cr 14.0 Co 8.0 Mo 3.5 Nb 3.5 Al 3.5 Ti 2.5	Precipitation hardened	650	860	—	—	—	—	—	—	—	—
Udimet 500	Ni 53.7 Cr 18.0 Co 18.5 Mo 4.0 Al 2.9 Ti 2.9	Precipitation hardened	650	760	—	—	—	—	—	—	—	—
			760	325	—	—	—	—	—	—	—	—
			870	125	—	—	—	—	—	—	—	—
Udimet 700	Ni 55.5 Cr 15.0 Co 17.0 Mo 5.0 Al 4.0 Ti 3.5	Precipitation hardened	650	705	—	—	—	—	—	—	—	—
			760	425	—	—	—	—	—	—	—	—
			870	200	—	—	—	—	—	—	—	—
			980	55	—	—	—	—	—	—	—	—
Waspaloy	Ni 58.7 Cr 19.5 Co 13.5 Mo 4.3 Al 1.3 Ti 3.0	Precipitation hardened	650	615	—	—	—	—	—	—	—	—
			760	290	—	—	—	—	—	—	—	—
			870	110	—	—	—	—	—	—	—	—

* Registered trade mark.

BIBLIOGRAPHY

1. INCO Alloys International Ltd., Product Handbook, 1990.
2. G.W. Meetham (ed.), 'The Development of Gas Turbine Materials', Applied Science Publishers Ltd, Barking, Essex, 1981.
3. *ASM Handbook*, 10th edition, vols. 1 and 2, ASM International, Materials Park, OH, 1990.

Table 22.32 CAST NICKEL ALLOYS, TENSILE PROPERTIES

Alloy	*Nominal composition wt. %, balance* Ni		*Test temperature* °C	*0.2% Yield strength* MPa	*Ultimate tensile strength* MPa	*Elongation* %
Inconel 713C	Cr	12.5	21	740	850	8
	Mo	4.2	538	705	860	10
	Nb	2.0	1 038		255	19
	Al	6.1				
	Ti	0.8				
	C	0.12				
B-1900	Cr	8.0	21	825	970	8
	Co	10.0	538	870	1 005	7
	Mo	6.0	1 093	195	270	11
	Ta	4.3				
	Al	6.0				
	Ti	1.0				
Inconel 625	Cr	21.5	21	350	710	48
	Mo	8.5	538	235	510	50
	Nb	4.0	1 093			
	Al	0.2				
	Ti	0.2				
	Fe	2.5				
Inconel 718	Cr	18.5	21	915	1 090	11
	Mo	3.0	538			
	Nb	5.1	1 093			
	Al	0.5				
	Ti	0.9				
	Fe	18.5				
Inconel 100	Cr	10.0	21	850	1 018	9
	Co	15.0	538	885	1 090	9
	Mo	3.0	870	695	885	6
	Al	5.5	1 038	283	441	6
	Ti	4.7				
	V	1.0				
Inconel 738	Cr	16.0	21	950	1 095	6
	Co	8.5	732	841	1 000	3
	Mo	1.75	982	345	455	10
	W	2.6				
	Ta	1.75				
	Nb	0.9				
	Al	3.4				
	Ti	3.4				
MAR-M 200	Cr	9.0	21	840	930	7
	Co	10.0	538	880	945	5
	W	12.5	1 093	286	325	6
	Nb	1.8				
	Al	5.0				
	Ti	2.0				
MAR-M 246	Cr	9.0	21	860	965	5
	Co	10.0	538	860	1 000	5
	Mo	2.5	1 066	234	379	11
	W	10.0	1 093	138	207	5
	Ta	1.5				
	Al	5.5				
	Ti	1.5				
Nimocast 75	Cr	20.0	21	179	500	39
	Ti	0.5				
	C	0.12				
Nimocast 80	Cr	19.5	21	520	730	15
	Al	1.4				
	Ti	2.3				
	Fe	1.5				
	C	0.05				
Nimocast 90	Cr	19.5	21	520	700	14
	Co	18.0	538	420	595	15
	Al	1.4	1 000	25	53	
	Ti	2.4				
	Fe	1.5				
	C	0.06				

(*continued*)

Table 22.32 CAST NICKEL ALLOYS, TENSILE PROPERTIES—*continued*

Alloy	*Nominal composition* *wt. %, balance* Ni		*Test temperature* °C	*0.2% Yield strength* MPa	*Ultimate tensile strength* MPa	*Elongation* %
Rene 80	Cr	14.0	21	848	1 030	5
	Co	9.5	538	730	1 030	8
	Mo	4.0	1 093		180	13
	W	4.0				
	Al	3.0				
	Ti	5.0				
Udimet 500	Cr	18.5	21	815	930	13
	Co	16.5	538	725	895	13
	Mo	3.5				
	Al	3.0				
	Ti	3.0				
MAR-M 002 + Hf	Cr	9.0	21	825	1 035	7
	Co	10.0	538	860	1 035	5
	Ta	2.5	1 093	345	550	12
	Al	5.5				
	Ti	1.5				
	Hf	1.5				

Table 22.33 CAST NICKEL ALLOYS, CREEP PROPERTIES

Alloy	*Nominal composition* *wt. %, balance* Ni		*Rupture strength,* MPa 100 hr	1 000 hr	*Test temperature* °C
Inconel 713C	Cr	12.5	370	305	815
	Mo	4.2	305	215	870
	Nb	2.0	130	70	980
	Al	6.1			
	Ti	0.8			
	C	0.12			
B-1900	Cr	8.0	510	380	815
	Co	10.0	385	250	870
	Mo	6.0	180	110	980
	Ta	4.3			
	Al	6.0			
	Ti	1.0			
Inconel 100	Cr	10.0	455	365	815
	Co	15.0	360	260	870
	Mo	3.0	160	90	980
	Al	5.5			
	Ti	4.7			
	V	1.0			
Inconel 738C	Cr	16.0	430	345	815
	Co	8.5	295	235	870
	Mo	1.75	140	90	980
	W	2.6			
	Ta	1.75			
	Nb	0.9			
	Al	3.4			
	Ti	3.4			
Inconel 738	Cr	16.0	462	338	815
	Co	8.5	324	228	870
	Mo	1.75	131	83	980
	W	2.6			
	Ta	1.75			
	Nb	0.9			
	Al	3.4			
	Ti	3.4			
MAR-M 200	Cr	9.0	495	415	815
	Co	10.0	385	295	870
	W	12.5	170	125	980
	Nb	1.8			
	Al	5.0			
	Ti	2.0			

(*continued*)

Table 22.33 CAST NICKEL ALLOYS, CREEP PROPERTIES—*continued*

Alloy	*Nominal composition wt. %, balance* Ni		*Rupture strength,* MPa 100 hr	1 000 hr	*Test temperature* °C
MAR-M 246	Cr	9.0	525	435	815
	Co	10.0	440	290	870
	Mo	2.5	195	125	980
	W	10.0			
	Ta	1.5			
	Al	5.5			
	Ti	1.5			
Nimocast 90	Cr	19.5	160	110	815
	Co	18.0	125	83	870
	Al	1.4			
	Ti	2.4			
	Fe	1.5			
	C	0.06			
Rene 80	Cr	14.0	620	496	760
	Co	9.5	350	240	870
	Mo	4.0	160	105	980
	W	4.0			
	Al	3.0			
	Ti	5.0			
Udimet 500	Cr	18.5	330	240	815
	Co	16.5	230	165	870
	Mo	3.5	90		980
	Al	3.0			
	Ti	3.0			

22.5.1 Directionally solidified and single crystal cast superalloys

Directionally solidified (DS) and single crystal (SC or SX) nickel-base superalloys were developed to increase the creep and thermal fatigue strength of first stage turbine blades in aero-engine gas turbines, and are now increasingly used in land-based gas turbines. Property improvements over those of polycrystalline superalloys result from the elimination of grain boundaries transverse to the principal stress axis and the alignment of the low-modulus <100> crystal direction with the stress axis. In the case of the SC alloys, further improvements were realised by optimising alloy chemistry in the absence of grain-boundary strengtheners such as C, B, and Zr.

Table 22.34 lists nominal compositions for common DS alloys; the so-called 'first-generation' alloys represent slight modifications from polycrystalline cast superalloy compositions (generally addition of Hf). The second-generation alloys were specifically designed for directional solidification and several incorporate Re to achieve creep strengths equivalent to first-generation single crystal alloys. Table 22.35 lists nominal compositions for common SC alloys. As mentioned above, the first-generation alloys eliminated the grain boundary strengtheners C, B, and Zr. Increasing levels of Re in the second and third generation alloys have led to even higher creep resistance, due to the potent solid-solution strengthening effect of Re at high temperatures.

Property data for the various alloys are not listed due to the proprietary and specialised nature of these alloys. Users should consult the developer or supplier for property data for their specific needs.

BIBLIOGRAPHY

1. K. Harris, G. L. Erickson and R. E. Schwer, pp. 994–1006 in 'ASM Handbook, vol. 1, 10th edition', ASM International, Materials Park, OH. 1990.
2. A. D. Cetel and D. N. Duhl, pp. 287–296 in 'Superalloys 1992' (eds. S.D. Antolovich, *et al.*), TMS, Warrendale, PA, 1992.
3. K. Harris, et al., pp. 297–306 in 'Superalloys 1992' (eds. S. D. Antolovich, *et al.*), TMS, Warrendale, PA. 1992.
4. E. W. Ross and K. S. O'Hara, pp. 257–258 in 'Superalloys 1992' (eds. S. D. Antolovich, *et al.*), TMS, Warrendale, PA, 1992.
5. E. Fleury and L. Remy, *Mater. Sci. Eng. A*, 1993, **A167**, 23.
6. G. L. Erickson, pp. 35–44 in 'Superalloys 1996' (eds. R.D. Kissinger, *et al.*), TMS, Warrendale, PA, 1996.
7. D. N. Duhl and M. L. Gell, United Technologies Corporation, G. B. Patent 21121812A–PWA 1483 alloy.
8. W. S. Walston, *et al.*, pp. 27–34 in 'Superalloys 1996' (eds. R. D. Kissinger, *et al.*), TMS, Warrendale, PA, 1996.

Table 22.34 SOME COMMON DIRECTIONALLY SOLIDIFIED NICKEL-BASE SUPERALLOYS

		Nominal composition, wt. %													
Alloy	*Developer/Producer*	Cr	Co	Mo	W	Re	Al	Ti	Nb	Ta	Hf	C	B	Zr	Ni	*Reference*
First-Generation																
MAR-M 002	Martin Metals	8	10	—	10	—	5.5	1.5	—	2.6	1.5	0.15	0.015	0.03	bal.	1
MAR-M 200 Hf	Martin Metals	8	9	—	12	—	5.0	1.9	1	—	2.0	0.13	0.015	0.03	bal.	1
MAR-M 247	Martin Metals	8	10	0.6	10	—	5.5	1.0	—	3	1.5	0.15	0.015	0.03	bal.	1
PWA 1422	Pratt and Whitney	10	10	—	12	—	5.0	2.0	1	—	1.5	0.14	0.015	0.1	bal.	2
Rene 80H	General Electric	14	9	4.0	4	—	3.0	4.7	—	—	0.8	0.16	0.015	0.01	bal.	1
Second-Generation																
CM 186 LC	Cannon-Muskegon	6	9	0.5	8	3	5.7	0.7	—	3	1.4	0.07	0.015	0.005	bal.	3
CM 247 LC	Cannon-Muskegon	8	9	0.5	10	—	5.6	0.7	—	3.2	1.4	0.07	0.015	0.01	bal.	1
PWA 1426	Pratt and Whitney	6.5	10	1.7	6.5	3	6.0	—	—	4	1.5	0.1	0.015	0.1	bal.	2
Rene 142	General Electric	6.8	12	1.5	4.9	2.8	6.2	—	—	6.4	1.5	0.12	0.015	0.02	bal.	4
Rene 150	General Electric	5	12	1.0	5	3	5.5	—	(2.2 V)	6	1.5	0.12	0.015	0.02	bal.	4

Table 22.35 SOME COMMON SINGLE CRYSTAL NICKEL-BASE SUPERALLOYS

		Nominal composition, wt. %														
Alloy	*Developer/Producer*	Cr	Co	Mo	W	Re	Al	Ti	V	Nb	Ta	Hf	C	B	Y	Ni	*Reference*
First-Generation																	
AM1	ONERA, France	8	6	2.0	6	—	5.2	1.2		—	8	—	—	—	—	bal.	5
CMSX-2	Cannon-Muskegon	8	5	0.6	8	—	5.6	1.0	—	—	6	—	—	—	—	bal.	6
CMSX-3	Cannon-Muskegon	8	5	0.6	8	—	5.6	1.0	—	—	6	0.1	—	—	—	bal.	6
CMSX-6	Cannon-Muskegon	10	5	3.0	—	—	4.8	4.7	—	—	2	0.1	—	—	—	bal.	6
CMSX-11B	Cannon-Muskegon	12.5	7	0.5	5	—	3.6	4.2	—	0.1	5	0.04	—	—	—	bal.	6
CMSX-11C	Cannon-Muskegon	15	3	0.4	4.5	—	3.4	4.2	—	0.1	5	0.04	—	—	—	bal.	6
PWA 1480	Pratt and Whitney	10	5	—	4	—	5.0	1.5	—	—	12	—	—	—	—	bal.	1
PWA 1483	Pratt and Whitney	13	9	2.0	4	—	3.6	4.0	—	—	4	—	—	—	—	bal.	7
Rene N4	General Electric	9	8	2.0	6	—	3.7	4.2	—	0.5	4	—	—	—	—	bal.	1
RR 2000	Rolls-Royce	10	15	3.0	—	—	5.5	4.0	1.0	—	—	—	—	—	—	bal.	1
SRR 99	Rolls-Royce	8	5	—	10	—	5.5	2.2	—	—	3	—	—	—	—	bal.	1
Second-Generation																	
CMSX-4	Cannon-Muskegon	6.5	9	0.6	6	3	5.6	1.0	—	—	6.5	0.1	—	—	—	bal.	6
PWA 1484	Pratt and Whitney	5	10	2.0	6	3	5.6	—	—	—	9	0.1	—	—	—	bal.	1
Rene N5	General Electric	7	7.5	1.5	5	3	6.2	—	—	—	6.5	0.15	0.05	0.004	0.01	bal.	8
SC 180	Garrett	5.3	10	1.7	5	3	5.2	1.0	—	—	8.5	0.1	—	—	—	bal.	6
Third-Generation																	
CMSX-10	Cannon-Muskegon	2	3	0.4	5	6	5.7	0.2	—	0.1	8	0.03	—	—	—	bal.	6
Rene N6	General Electric	4.2	12.5	1.4	6	5.4	5.7	—	—	—	7.2	0.15	0.05	0.004	0.01	bal.	8

22.6 Mechanical properties of titanium and titanium alloys

Table 22.36 TITANIUM AND TITANIUM ALLOYS. CORRESPONDING GRADES OR SPECIFICATIONS

Note for 8th edition: In the case of the UK, France, and Germany, this table refers to designations and standards that may have been superceded by European (EN) and international (ISO) standards. This table is retained for information only.

IMI designation	*UK British Standards (Aerospace series) and Min. of Def.* DTD series*	*France* AIR-9182, 9183, 9184	*Germany BWB series*†	*AECMA recom-mendations*	*USA AMS series*‡	*USA ASTM series*
IMI 115	BS TA 1, DTD 5013	T-35	3.7024	Ti-POI		ASTM grade 1
IMI 125	BS TA 2,3,4,5	T-40	3.7034	Ti-PO2	AMS 4902, 4941, 4942, 4951	ASTM grade 2
IMI 130	DTD 5023, 5273, 5283, 5293	T-50			AMS 4900	ASTM grade 3
IMI 155	BS TA 6	T-60	3.7064	Ti-PO4	AMS 4901	ASTM grade 4
IMI 160	BS TA 7,8,9				AMS 4921	
IMI 230	BS TA 21, 22, 23, 24, BS TA 52–55, 58	T-U2	3.7124	Ti-P11		
IMI 315	DTD 5043					
IMI 317	BS TA 14, 15, 16, 17	T-A5E		Ti-P65	AMS 4909, 4910, 4926, 4924, 4953, 4966	
IMI 318	BS TA 10, 11, 12, 13, 28, 56	T-A6V	3.7164	Ti-P63	AMS 4911, 4928, 4934, 4935, 4954, 4965, 4967	ASTM grade 5
IMI 318 ELI (extra low interstitial)	—	T-A6VELI			AMS 4907, 4930, 4931	ASTM grade 3, F. 136
IMI 325	—	T-A3V2.5	3.7194		AMS 4943 4944	ASTM grade 9
IMI 550	BS TA 45–51, 57	T-A4DE	3.7184	Ti-P68		
IMI 551	BS TA 39–42					
IMI 624	—	T-A6Zr4 DE	3.7144		AMS 4919, 4975, 4976	
IMI 646	—	—	—	—	AMS 4981	
IMI 662	—	—	3.7174	—	AMS 4918, 4971, 4978, 4979	
IMI 679	BS TA 18–20, 25–27				AMS 4974	
IMI 680	DTD 5213	T-E11DA				
IMI 685	BS TA 43, 44	T-A6ZD	3.7154	Ti-P67		
IMI 811	—	T-A8DV			AMS 4915, 4916	
IMI 834	—	T-A6E Zr4Nb				

* UK BS 3531 Part 1 (Metal Implants in Bone Surgery), and Draft British Standard for Lining of Vessels and Equipment for Chemical Processes, Part 9, also refer.

† Germany DIN 17850, 17860, 17862, 17863, 17864 (3.7025/35/55/65), and TUV 230-1-68 Group, I II, III and IV also refer.

‡ USA MIL-T-9011, 9046, 9047, 14577, 46038, 46077, 05-10737 and ASTM B265-69, B338-65, B348-59T, B367-61T. B381-61T, B382-61T also refer.

Table 22.37 PURE TITANIUM, TYPICAL MECHANICAL PROPERTIES AT ROOM TEMPERATURE

*Designation**	*Grade*	*Condition*	0.2 % *proof stress* MPa	*Tensile strength* MPa	*Elongation %* *on* 50 mm	*on* 5D	*Red in area* %	*Specification bend radius* 180° bend <1.83 mm	<3.25 mm	*Mod. of elasticity* GPa	*Mod. of rigidity* GPa
Iodide	Pure, 60 HV		103	241	55		80				
IMI 115	Commercially pure	Annealed sheet	255	370	33			1t	2t		
		Annealed rod	220	370		40	70				
		Annealed wire		390	38						
IMI 125	Commercially pure	†Annealed sheet	340	460	30			$1\frac{1}{2}$t	2t		
		Annealed rod	305	460		28	57				
		Annealed tube	325	480	35						
IMI 130	Commercially pure	Annealed sheet	420	540	25			2t+	$2\frac{1}{2}$ t		
		Annealed rod	360	540		24	48			105	38
		Annealed wire		550	24						
		Hard-drawn wire		700	11.5						
IMI 155	Commercially pure	Annealed sheet	540	640	24			$2\frac{1}{2}$ t	3t		
IMI 160	Commercially pure	Annealed rod	500	670		23	46				
		Annealed wire		690	24						

* IMI Nomenclature. † Up to 16.3 mm.

Table 22.38 TITANIUM ALLOYS TYPICAL MECHANICAL PROPERTIES AT ROOM TEMPERATURE

*Designation**	*Nominal composition* %		*Condition*	*0.2% proof stress* MPa	*Tensile strength* MPa	*Elongation %* *on* 50 mm	*on* 5D	*Red. in area* %	*Specification bend radius* 180°	*Mod. of elasticity* GPa	*Mod. of rigidity* GPa
IMI 230	Cu	6.0	Annealed sheet	520	620	24			2 t (0.5–3 mm)	125	
			Aged sheet	670	770	20			2t (typical)		
			Annealed rod	500	630		27	45			
			Aged rod	580	740		22	41		125	
IMI 260	Pd	0.2	Similar to commercially Pure Titanium 115								
IMI 261	Pd	0.2	Similar to commercially Pure Titanium 125								
IMI 315	Al	2.0	Annealed rod	590	720		21	50		120	
	Mn	2.0									
IMI 317	Al	5.0	Annealed sheet	820	860	16			4 t (<2 mm)		
	Sn	2.5							$4\frac{1}{2}$ t (≤3 mm)		
			Annealed rod	930	1 000		15	37		120	

(*continued*)

Table 22.38 TITANIUM ALLOYS TYPICAL MECHANICAL PROPERTIES AT ROOM TEMPERATURE—*continued*

*Designation**	*Nominal composition* %		*Condition*	*0.2% proof stress* MPa	*Tensile strength* MPa	*Elongation %* *on* 50 mm	*on* 5D	*Red. in area* %	*Specification bend radius* 180°	*Mod. of elasticity* GPa	*Mod. of rigidity* GPa
IMI 318	Al	6.0	Annealed sheet	1 110	1 160	10			5 t (≤3.25 mm)		
	V	4.0	Annealed rod	990	1 050		15	40		106	46
			Aged rod (fastener stock)	1 050	1 140		15	40			
			Hard-drawn wire		1 410	4					
IMI 550	Al	4.0	F.h.t. rod	1 070	1 200		14	42		116	
	Mo	4.0									
	Sn	2.0									
	Si	0.5									
IMI 551	Al	4.0	F.h.t.rod	1 140	1 300		12	40		113	43
	Mo	4.0									
	Sn	4.0									
	Si	0.5									
IMI 679	Sn	11.0	Quenched and aged rod	1 080	1 230		11	40		108	46
	Al	2.25	Air-cooled and	1 000	1 120		13	45			
	Mo	1.0	aged rod								
	Si	0.2									
IMI 680	Sn	11.0	Quenched and aged	1 200	1 350		12	37		115	
	Mo	4.0	rod								
	Al	2.25	Furnance-cooled and	1 080	1 160		14	47			
	Si	0.2	aged rod								
IMI 685	Al	6.0	F.h.t. rod	920	1 020		11	22		124	47
	Zr	5.0									
	Mo	0.5									
	Si	0.25									
IMI 829	Al	5.5	F.h.t. rod	848	965		12	22		120	
	Sn	3.5									
	Zr	3.0									
	Nb	1.0									
	Mo	0.3									
	Si	0.3									
IMI 834	Al	5.8	F.h.t. rod	931	1 067		13	22		120	
	Sn	4.0									
	Zr	3.5									
	Nb	0.7									
	Mo	0.5									
	Si	0.35									
	C	0.06									

* IMI Nomenclature.

Table 22.39 COMMERCIALLY PURE TITANIUM SHEET, TYPICAL VARIATION OF PROPERTIES WITH TEMPERATURE

*Designation**	*Temperature* °C	0.2% *proof stress* MPa	*Tensile strength* MPa	*Elongation on* 50 mm %	*Mod. of elasticity* GPa	*Transformation temperature* °C
IMI 115	−196	442	641	34		$\alpha/\alpha+\beta$
	−100	306	444	34		865
	20	207	337	40		
	100	168	296	43		
	200	99	218	38		
	300	53	167	47		
	400	42	131	52		
	450	36	120	49		
IMI 125	20	334	479	31		
	100	250	397	32		
	200	184	300	40		
	300	142	232	45		
	400	127	190	38		
	450	119	175	35		
IMI 130	−196	730	855	28		
	−100	590	737	28		
	20	394	547	28	108	
	100	315	462	29	99	$\alpha+\beta/\beta$
	200	205	331	37	91	915
	300	139	247	40	83	
	400	102	199	34	65	
	450	93	182	28		
	500				46	
IMI 155	20	460	625	25		
	100	372	537	26		
	200	219	386	32		
	300	151	281	36		
	400	110	221	33		
	450	96	202	26		

* IMI nomenclature.

Table 22.40 TITANIUM ALLOYS, TYPICAL VARIATION OF PROPERTIES WITH TEMPERATURE

Designation	*Nominal composition* %		*Condition*	*Temperature* °C	0.2% *proof stress* MPa	*Tensile strength* MPa	*Elongation* % *on* 50 mm	*on* 5D	*Red. in area* %	*Mod. of elasticity* GPa	*Transformation temperature* °C
IMI 230	Cu	2.5	S.h.t. (trans.)	20	500	605	24				$\alpha/\alpha+\beta$
				100	410	540	29				790
				200	310	450	33				
				300	270	410	31				$\alpha+\beta/\beta$
				400	250	380	30				895 ± 10
				500	220	380	33				
			Aged sheet (trans.)	20	622	761	24				
				100	553	704	23				
				200	471	635	26				
				300	457	607	23				
				400	429	573	19				
				500	357	468	21				
			Aged	20	638	795		22	40	107	
				100	601	761		21	39	100	
				200	507	687		23	45	92	
				300	496	658		20	50	85	
				400	415	592		21	53	78	
				500	361	491		27	57	71	
IMI 260	Pd	0.2	Similar to IMI 115								
IMI 262	Pd	0.2	Similar to IMI 125								
IMI 315	Al	2.0	Annealed rod	20	618	757		18	41	110	$\alpha+\beta/\beta$
	Mn	2.0		100	510	649		21	46	107	915 ± 20
				200	386	525		22	48	97	
				300	293	432		19	50	86	
				400	278	417		18	56	76	
				500	201	340		22	72	62	
IMI 317	Al	5.0	Annealed rod	20	822	919		18	39	112	$\alpha/\alpha+\beta$
	Sn	2.5		100	692	798		19	40	109	950
				200	494	638		18	44	105	
				300	415	576		19	42	89	$\alpha+\beta/\beta$
				400	374	522		18	41	84	1025 ± 20
				500	346	485		21	57	81	

IMI 318	Al	6.0	Annealed rod	−196	1 560	1 675	6	29		$\alpha+\beta/\beta$
	V	4.0		−100	1 165	1 265	12	33		1 000 ± 15
				20	970	1 040	15	38	106	
				100	825	920	17	43	102	
				200	710	815	18	49	96	
				300	645	750	18	56	90	
				400	580	700	18	63	85	
				500	450	605	26	72	79	
				600	125	265	58	85		
				700	40	135	127	94		
			Heat-treated rod	20	1 035	1 145	14			
			(fastener stock)	100	925	1 035	15			
				200	805	925	16			
				300	710	850	16			
				400	635	805	18			
				500	540	695	25			
IMI 550	Al	4.0	F.h.t. rod	20	1 081	1 220	15	49	116	$\alpha+\beta/\beta$
	Mo	4.0		100	965	1 130	15	49	112	980 ± 10
	Sn	2.0		200	805	960	16	60	106	
	Si	0.5		300	700	900	16	55	101	
				400	655	835	17	60	95	
				500	585	780	19	68	90	
				600	310	585	26	83	85	
IMI 551	Al	4.0	F.h.t. rod	20	1 250	1 390	10	27	113	$\alpha+\beta/\beta$
	Mo	4.0		100	1 125	1 300	11	29	108	1 050 ± 15
	Sn	4.0		200	925	1 145	14	38	103	
	Si	0.5		300	815	1 045	15	38	98	
				400	745	970	14	41	93	
				500	670	920	18	55	88	
				600	460	755	27	65	81	
IMI 679	Sn	11.0	Quenched and	20	1 050	1 230	10	37		$\alpha+\beta/\beta$
	Zr	5.0	aged rod	100	940	1 145	11	43		950 ± 10
	Al	2.25		200	820	1 020	12	45		
	Mo	1.0		300	740	990	11	46		
	Si	0.2		400	710	940	11	46		
				450	680	910	11	46		
			Air-cooled and	20	1 020	1 095	14	41	108	
			aged rod	100	895	995	16	47	103	
				200	770	900	16	49	99	
				300	695	865	14	49	94	
				400	665	850	14	48	90	
				500	600	795	15	48	85	

(*continued*)

Table 22.40 TITANIUM ALLOYS, TYPICAL VARIATION OF PROPERTIES WITH TEMPERATURE—*continued*

Designation	*Nominal composition* %		*Condition*	*Temperature* °C	0.2% *proof stress* MPa	*Tensile strength* MPa	*Elongation* % *on* 50 mm	*on* 5D	*Red. in area* %	*Mod. of elasticity* GPa	*Transformation temperature* °C
IMI 680	Sn	11.0	Quenched and	20	1 180	1 330		12	43	106	$\alpha + \beta/\beta$
	Mo	4.0	aged rod	100	1 020	1 190		14	49	100	945 ± 15
	Al	2.25		200	905	1 105		15	53	96	
	si	0.2		300	835	1 075		15	56	94	
				400	805	1 020		14	57	90	
				450	725	975		13	54	88	
			Furnace-cooled	−196	1 630	1 730		$8\frac{1}{2}$	36		
			and aged rod	−100	1 280	1 380		10	43		
				20	1 030	1 130		15	49		
IMI 685	Al	6.0	F.h.t. rod	−196	1 480	1 560		6	13		$\alpha + \beta/\beta$
	Zr	5.0		−100	1 140	1 270		10	18		1 020 ± 10
	Mn	0.5		20	890	1 030		12	22	124	
	Si	0.25		100	800	935		13	22	120	
				200	720	850		15	24	114	
				300	650	800		16	27	108	
				400	595	750		18	31	102	
				500	535	695		19	37	95	
IMI 829	Al	5.5	F.h.t. rod	20	895	1 028		$10\frac{1}{2}$	22	119	$\alpha + \beta\beta$
	Sm	3.5		200	622	792		$14\frac{1}{2}$	28	110	1 015 ± 15
	Zr	3.0		500	501	665		15	36	93	
	Nb	1.0		540	487	653		16	42	91	
	Mo	0.3		600	457	634		14	38	88	
	Si	0.3									
IMI 834	Al	5.8	F.h.t. rod	20	931	1 067		13	22	120	1 045
	Sn	4.0		100	840	962		13	23	116	
	Zr	3.5		200	746	885		14	27	112	
	Nb	0.7		300	700	832		14	32	106	
	Mo	0.5		400	662	790		14	36	102	
	Si	0.35		500	609	764		15	42	96	
	C	0.06		600	505	656		16	50	92	

Table 22.41 COMMERCIALLY PURE TITANIUM—TYPICAL CREEP PROPERTIES

IMI designation	*Temperature* °C	*Stress* MPa *to produce* 0.1% *plastic strain in* 1 000 h	10 000 h	100 000 h
IMI 130	20	288	270	207
	50	243	221	165
	100	179	165	119
	150	140	133	96
	200	113	116	77
	250	96	101	66
	300	87	83	55
IMI 155	20	309	278	260
	50	252	232	213
	100	188	170	157
	150	145	131	122
	200	116	108	104
	250	102	97	94
	300	93	90	86

Table 22.42 TITANIUM ALLOYS—TYPICAL CREEP PROPERTIES

IMI designation	*Nominal composition* %		*Condition*	*Temperature* °C	*Stress* MPa *to produce* 0.1% *total plastic strain in* 100 h	300 h	500 h	1 000 h
IMI 230	Cu	2.5	Aged sheet	200	435	—	—	—
				300	375	—	—	—
				400	220	—	—	—
				450	109	—	—	—
			Annealed sheet	20	360	—	—	—
				100	279	—	—	—
				200	235	—	—	—
				300	202	—	—	—
				400	125	—	—	—
IMI 317	Al	5.0	Annealed rod	20	633	608	—	593
	Sn	2.5		100	474	463	—	458
				200	370	—	—	370
				300	359	—	—	359
				400	337	—	—	337
				500	162	119	—	88
IMI 318	Al	60	Annealed rod	20	832	818	—	788
	V	4.0		100	704	680	—	676
				200	638	636	—	635
				300	576	568	—	—
				400	287	144	—	102
				500	32	18	—	—
IMI 550	Al	4.0	Fully heat-	300	724	718	—	710
	Mo	4.0	treated bar	400	551	516	—	471
	Sn	2.0		450	254	174	—	101
	Si	0.5		500	82	51	—	31
IMI 551	Al	4.0	Fully heat-	400	621	575	540	501
	Mo	4.0	treated rod	450	307	217	—	—
	Sn	4.0						
IMI 679	Sn	11.0	Air-cooled and	20	896	880	—	880
	Zr	5.0	aged rod	150	703	695	—	672
	Al	2.25		300	664	664	—	649
	Mo	1.0		400	579	571	—	526
	Si	0.2		450	448	386	—	247
				500	131	93	—	62

(*continued*)

Table 22.42 TITANIUM ALLOYS—TYPICAL CREEP PROPERTIES—*continued*

IMI designation	*Nominal composition* %	*Condition*	*Temperature* °C	*Stress MPa to produce 0.1% total plastic strain in* 100 h	300 h	500 h	1 000 h
IMI 680	Sn 11.0 Mo 4.0 Al 2.25 Si 0.2	Quenched and aged rod	20 150 200 300 400 450 500	1 127 945 862 804 555 298 88	1 112 942 856 788 540 209 51	— — — — — — —	— — — — — — —
		Furnace-cooled and aged rod	300 350 400	570 540 490	— — —	— — —	— — —
IMI 685	Al 6.0 Zr 5.0 Mo 0.5 Si 0.25	Heat-treated forgings	200 300 400 450 500	599 551 497 461 408	— — — — —	592 541 480 431 340	589 535 462 426 —
IMI 829	Al 5.5 Sn 3.5 Zr 3.0 Nb 1.0 Mo 0.3 Si 0.3	Fully heat treated rod	450 500 550 600	478 420 300 130	— — — —	— — — —	— — — —
IMI 834	Al 5.8 Sn 4.0 Zr 3.5 Nb 0.7 Mo 0.5 Si 0.35 C 0.06	Heat-treated forgings	500 550 600	461 339 205	— — —	— — —	— — —

Table 22.43 TITANIUM AND TITANIUM ALLOYS—TYPICAL FATIGUE PROPERTIES

IMI designation	*Nominal composition* %	*Condition*	*Temperature* °C	*Tensile strength* MPa	*Details of test*	*Endurance limit for* 10^7 *cycles stated* MPa
					Rotating bend	
IMI 115	Commercial purity	Annealed rod	Room	354	Smooth $K_t = 1$	±193
				354	Notched $K_t = 3$	±123
					Rotating bend	
IMI 125	Commercial purity	Annealed rod	Room	417	Smooth $K_t = 1$	±232
				417	Notched $K_t = 3$	±154
					Rotating bend	
IMI 130	Commercial purity	Annealed rod	Room	550	Smooth $K_t = 1$	±270
				550	Notched $K_t = 2$	±170
				550	Notched $K_t = 3.3$	±170
					Direct stress (Zero mean)	
				550	Smooth $K_t = 1$	±263
				550	Notched $K_t = 1,5$	±247
				550	Notched $K_t = 2$	±170
				550	Notched $K_t = 3.3$	±116
				589	Smooth $K_t = 1$	±278
				589	Notched $K_t = 2$	±147
				589	Notched $K_t = 3$	±123
				589	Notched $K_t = 4$	±116

(*continued*)

Table 22.43 TITANIUM AND TITANIUM ALLOYS—TYPICAL FATIGUE PROPERTIES—*continued*

IMI designation	*Nominal composition %*	*Condition*	*Temperature °C*	*Tensile strength MPa*	*Details of test*	*Endurance limit for 10^7 cycles stated MPa*
					Direct stress (Zero mean)	
IMI 160	Commercial purity	Annealed rod	Room	674	Smooth K_t	±376
IMI 230	Cu 2.5	Annealed sheet	Room	564	Reversed bend	±390
		Aged sheet	room	772	Reversed bend	±490
					Direct stress (Zero minimum)	
		Aged sheet	room	761	Smooth $K_t = 1$	0→560
					Rotating bend	
		Annealed rod	room	598	Smooth $K_t = 1$	±370
			400		Smooth $K_t = 1$	±150
					Direct stress (Zero mean)	
		Annealed	Room	638	Smooth $K_t = 1$	±280
					Rotating bend	
		Aged rod	Room	700	Smooth $K_t = 1$	±450
			400	—	Smooth $K_t = 1$	±290
					Direct stress (Zero mean)	
		Aged rod	Room	792	Smooth $K_t = 1$	±470
					Notched $K_t = 3.3$	±200
IMI 260	Pd 0.2	Similar to IMI 115				
IMI 262	Pd 0.2	Similar to IMI 125				
					Rotating bend	Limits for this alloy 10^8 cycles
IMI 317	Al 5.0 Sn 2.5	Annealed rod	Room	—	Smooth $K_t = 1.0$	±371
					Notched $K_t = 2.0$	±263
					Notched $K_t = 3.3$	±239
					Direct stress (Zero mean)	
					Smooth $K_t = 1.0$	±433
					Notched $K_t = 1.5$	±278
					Notched $K_t = 2.0$	±201
					Notched $K_t = 3.3$	±154
					Rotating bend	
IMI 318	Al 6.0 V 4.0	Annealed rod	Room	960	Smooth $K_t = 1$	±470
				960	Notched $K_t = 2.7$	±230
					Direct stress (Zero minimum)	
				1 015	Smooth $K_t = 1$	0→750
				1 015	Notched $K_t = 1$	0→325
					Direct stress (Zero minimum)	
IMI 550	Al 4.0 Mo 4.0 Sn 2.0 Si 0.5	Fully heat-treated rod	Room	1 180	Smooth $K_t = 1$	0→850
				1 180	Notched $K_t = 3$	0→350
					Rotating bend	
					Rotating bend	
					Smooth $K_t = 1$	±587
					Notched $K_t = 2.4$	±394

(*continued*)

Table 22.43 TITANIUM AND TITANIUM ALLOYS—TYPICAL FATIGUE PROPERTIES—*continued*

IMI designation	*Nominal composition* %		*Condition*	*Temperature* °C	*Tensile strength* MPa	*Details of test*	*Endurance limit for 10^7 cycles stated* MPa
						Rotating bend	
IMI 551	Al	4.0	Fully heat-	Room	—	Smooth $K_t = 1$	±750
	Mo	4.0	treated rod		—	Notched $K_t = 3.2$	±430
	Sn	4.0					
	Si	0.5					
						Rotating bend	
IMI 679	Sn	11.0	Air-cooled and	Room	—	Smooth $K_t = 1.0$	±641*
	Zr	5.0	aged rod	200	—	Smooth $K_t = 1.0$	±510*
	Al	2.25		400	—	Smooth $K_t = 1.0$	±510*
	Mo	1.0		450	—	Smooth $K_t = 1.0$	±556
	Si	0.2		500	—	Smooth $K_t = 1.0$	±495
						Rotating bend	(Limits for 2×10^7 cycles)
IMI 680	Sn	11.0	Quenched and	Room	1 272	Smooth $K_t = 1$	±710
	Mo	4.0	aged rod		1 272	Notched $K_t = 2$	±340
	Al	2.25			1 272	Notched $K_t = 3.3$	±293
	Si	0.2					
						Direct stress (Zero mean)	(Limits for 2×10^7 cycles)
				Room	1 272	Smooth $K_t = 1$	±695
						Notched $K_t = 2$	±371
						Notched $K_t = 3.3$	±232
						Rotating bend	(Limits for 10^8 cycles)
				Room	—	Smooth $K_t = 1$	±648
				200	—	Smooth $K_t = 1$	±495
				400	—	Smooth $K_t = 1$	±479
						Direct stress Zero	
			Furnace-cooled rod	Room	1 100	Smooth $K_t = 1$	±680
						Direct stress (Zero mean)	
IMI 685	Al	6.0	Fully heat-	20	—	Smooth $K_t = 1$	±440
	Zr	5.0	treated rod	450	—	Smooth $K_t = 1$	±300
	Mo	0.5		520	—	Smooth $K_t = 1$	±260
	Si	0.25					
						Direct stress (Zero minimum)	
				450	—	Smooth $K_t = 1$	0→475
				520	—	Smooth $K_t = 1$	0→425
						Direct stress (Zero minimum)	
			Fully heat-	Room	—	Smooth $K_t = 1$	0→640
			treated forging	Room	—	Notched $K_t = 3.5$	0→220
				475	—	Smooth $K_t = 1$	0→460
				475	—	Notched $K_t = 3.5$	0→210
						Direct stress (Zero minimum)	
IMI 829	Al	5.5	Fully heat-	Room	—	Smooth $K_t = 1$	0→550
	Sn	3.5	treated rod			Notched $K_t = 3$	0→260
	Zr	3.0					
	Nb	1.0					
	Mo	0.3					
	Si	0.3					
						Direct stress (Zero minimum)	
IMI 834	Al	5.8	Fully heat-	Room	—	Smooth $K_t = 1$	0→577
	Sn	4.0	treated rod		—	Notched $K_t = 2$	0→363
	Zr	3.5					
	Nb	0.7					
	Mo	0.5					
	Si	0.35					
	C	0.06					

* Limit for 10^8 cycles.

Table 22.44 IZOD IMPACT PROPERTIES OF TITANIUM AND TITANIUM ALLOYS

IMI designation	*Nominal composition %*	*Condition*	*Izod value* Joules(ft lbf)*							
			−196°C	*−78°C*	20°C	100°C	200°C	300°C	400°C	500°C
IMI 130†	Commercially pure	Annealed rod	—	62.4 (46)	61.0 (45)	62.4 (46)	72 (53)	82 (60½)	84 (62)	82 (62½)
IMI 317	Sn 5.0 Al 2.5	Annealed rod	17.6 (13)	20.3 (15)	27.1 (20)	35.2 (26)	52.8 (39)	63.7 (47)	70.5 (52)	71.8 (53)
IMI 318	Al 6.0 V 4.0	Annealed rod	13.5 (10)	14.9 (11)	20.3 (15)	25.7 (19)	40.6 (30)	65.0 (48)	83.5 (63)	92.0 (68)
IMI 550	Al 4.0 Mo 4.0 Sn 2.0 Si 0.5	Fully heat-treated rod	—	—	19.0 (14)	—	—	—	—	—

IMI designation	*Nominal composition %*	*Condition*	*Charpy value* Joules(ft lbf)							
			−196°C	*−78°C*	20°C	100°C	200°C	300°C	400°C	500°C
IMI 551	Al 4.0 Mo 4.0 Sn 4.0 Si 0.5	Fully heat-treated rod	13.5 (10)	19 (14)	20.3 (15)	21.7 (16)	24.4 (18)	26.5 (19½)	28.5 (21)	31.2 (23)
IMI 679	Sn 11.0 Zr 5.0 Al 2.25 Mo 1.0 Si 0.2	Air-cooled and aged	10.8 (8)	13.5 (10)	14.9 (11)	16.3 (12)	19 (14½)	25 (18½)	30 (22)	33.9 (25)
IMI 680	Sn 11.0 Mo 4.0 Al 2.25 Si 0.2	Quenched and aged rod	8.1 (6)	8.8 (6½)	10.8 (8)	12.2 (9)	14.9 (11)	17.6 (13)	20.3 (15)	25.7 (19)
IMI 685	Al 6.0 Zr 5.0 Mo 0.5 Si 0.25	Fully heat-treated rod	31.2 (23)	39.3 (29)	43.4 (32)	—	—	—	—	—

* BSS 131 (1) 0.45 in diameter straight notched test pieces. † Izod values of commercial purity titanium are appreciably affected by variation in hydrogen content within commercial limits (0.008% maximum) in Ti 130 rod.

22.7 Mechanical properties of zinc and zinc alloys

Table 22.45 MECHANICAL PROPERTIES OF ZINC ALLOYS AT ROOM TEMPERATURE

	Composition						*Properties*		
Zinc	Unalloyed zinc is generally used only in the wrought form. The data here are some typical values								
	Zn min	Pb	Cd max	Sn	Fe	*Total* Pb+ Cd + Sn + Fe + Cu (max)	*Parallel to rolling direction*	*Across rolling direction*	
BS 3436 Zn 1	99.99	0.003	0.003	0.001	—	0.01	Zn 1		
Zn 2	99.95	0.03	0.02	0.001	0.01	0.05	Tensile strength		
Zn 3	99.5	0.35	0.15	0.001	0.03	0.5	MPa	120	150
Zn 4	98.5	1.35	0.15	0.02	0.04	1.5	Elongation %	60–80	40–60
							Hardness Vickers	30	

(*continued*)

Table 22.45 MECHANICAL PROPERTIES OF ZINC ALLOYS AT ROOM TEMPERATURE—*continued*

	Composition			*Properties*		
Zinc-copper-titanium	Cu	Ti	Balance			
	0.14	0.1–0.15	zinc of 99.995% purity	Tensile strength MPa	180	216
				Elongation %	35	20
				Hardness (Brinell)	40–45	

These alloys are used principally in pressure die castings and other castings. Properties are given in Chapter 26 Casting and foundry data.

22.8 Mechanical properties of zirconium and zirconium alloys

Table 22.46 MECHANICAL PROPERTIES OF ZIRCONIUM ALLOYS AT ROOM TEMPERATURE

Material	*Nominal composition*	*Condition*	0.1% *proof stress* MPa	*UTS* MPa	*Elongation* %	*Macro-hardness* HV	*Reference*
Zr (ex iodide)	>99.9% purity Impurities in ppm by wt O_2 65, N_2 15, H_2 12, Hf 35, Ni 20	Crystal bar, cold rolled and vacuum annealed 2 h at 750°C	100–130	170–210	40–45 on 1 in gauge	85–100	1
Zr (ex sponge)	>99.6% purity Impurities in ppm by wt O_2 1300, N_2 80, H_2 20, Hf 400, Ni 40	Sheet material, cold rolled and vacuum annealed 2 h at 750°C	250–310 in r.d.	350–390 in r.d.	23–31 on 1 in gauge	195–215	1
Zircaloy 2	Sn 1.2–1.7% Fe 0.07–0.2% Cr 0.05–0.15% Ni 0.03–0.08% Zr–remainder	Plate materials cold rolled and annealed 1 h at 750°C	340† in r.d. 490 in t.d.	450 in r.d. 520 in t.d.	29 in r.d. 23 in t.d. on 1 in gauge	205–220*	2 1
Zirconium 30*	Cu 0.46–0.66% Mo 0.50–0.60% Zr–remainder	Sheet and strip	220–320 in r.d.	470–550 in r.d.	20–31 2 in gauge	130–180	3
Zr/2½% Nb	Nb 2.55% Impurities in ppm by wt O_2 1050, N_2 20, H_2 10, Hf 70 N < 40	Plate, hot rolled at 750°C later annealed ½ h at 700°C	470 for forged product	590 for forged product 410–500 ppm O_2 600–2 500 ppm O_2	24 gauge unspecified‡	150–500 ppm O_2 230–2 500 ppm O_2	4 3
Zr 702	Commercially pure with up to Hf 4.5	Annealed sheet	207	379	16	5T	
Zr 704	Cr + Fe 0.2–0.4 Sn 1–2	Annealed sheet	241	413	14	5T	
Zr 705	Nb 2.5 O_2 0.18	Annealed sheet	379	552	16	3T	
Zr 706	Nb 2.5 O_2 0.16	Annealed sheet	345	510	20	2½T	

* Imperial Metal Industries Nomenclature.

Table 22.47 MECHANICAL PROPERTIES OF ZIRCONIUM ALLOYS AT ELEVATED TEMPERATURE

Material	*Nominal composition*	*Test temperature* °C	0.1% *proof stress* MPa	*UTS* MPa	*Elongation* %	*Reference*
Zr (ex sponge)	>99.6% purity Impurities in ppm by wt O_2 1 300, N_2 80, H_2 20, Hf 400, Ni 40	371	50†	110	57	3
Zircalloy 2	Sn 1.2–1.7% Fe 0.07–0.2% Cr 0.05–0.15% Ni 0.03–0.08% Zr–remainder	300	92–126	210–260	33–44 on 2 in gauge	3
Zirconium 30	Cu 0.46–0.66% Mo 0.50–0.60% Zr–remainder	300	160	250	34 on $5.65\sqrt{A}$ gauge	3
Zr/2$\frac{1}{2}$% Nb	Nb 2.55% Impurities in ppm by wt O_2 1 050, N_2 20, H_2 10, Hf 70, Ni < 40	300	210	340 280–500 ppm O_2^* 350–2 500 ppm O_2^*	33 gauge unspecified	3 4*
Zr 702	Commercially pure with up to Hf 4.5	100 200 300 400	296 262 207 145	413 351 275 227	32 37 47 54	
Zr 704	Cr + Fe 0.2–0.4	100 200 300 400	344 289 241 193	510 489 434 358	22 23 27 32	
Zr 705	Nb 2.5 O_2 0.18	100 200 300 400	455 379 317 269	565 489 427 372	26 31 33 33	

r.d. rolling direction.
t.d. transverse direction.
*Imperial Metal Industries nomenclature.
†0.2% proof stress.

REFERENCES TO TABLES 22.46 AND 22.47

1. B. J. Gill, Dept. of Metallurgy and Materials Tech., University of Surrey, Guildford, England, 1972.
2. W. Evans and G. W. Parry, *Electrochem. Tech.*, 1966, **4**, 225.
3. Imperial Metal Industries pubn 2 Ed/MK105/33/366.
4. J. Winton and R. A. Murgatroyd, *Electrochem. Tech.*, 1966, **4**, 358.

22.9 Tin and its alloys

The applications of tin are governed very largely by its low melting point, excellent fluidity when molten, relative softness, formability and readiness to form alloys with other metals.

The low melting point of tin means that at normal ambient temperature the metal is nearly 60% of its melting point on the absolute Temperature Scale. This results in rather low mechanical strength at room temperature, as would be expected from any metal tested at a temperature corresponding to such a high proportion of its Absolute melting point, i.e. 505 K. Because of this, pure tin recrystallises readily at room temperature. Consequently, unlike the majority of industrial metals, only slight work hardening occurs initially in tin, followed by work-softening with further deformation due to grain growth. A similar behaviour is found in tin-rich alloys. The mechanical properties recorded for tin-rich alloys are strongly dependent on impurity levels and the strain rate.

Within reasonable limits any tin-base alloy can be die cast successfully. The choice of alloy, therefore, rests on such matters as mechanical properties, wear resistance or cost rather than on any consideration of the casting behaviour.

PURE TIN—MECHANICAL PROPERTIES

Tensile strength (at 0.4 mm/mm min) at	20°C	14.5 MPa
	100°C	11.0 N mm^{-2}
	200°C	4.5 N mm^{-2}
Shear strength at 20°C		12.3 MPa
Hardness at 20°C		3.9 HB
100°C		2.3 HB
200°C		0.9 HB
Young's modulus at 20°C		49.9 GPa
Rigidity modulus at 20°C		18.4 GPa
Poisson's ratio		0.357
Creep strength at 15°C		
(approx. life at 2.3 N mm^{-2})		170 days
(approx. life at 1.4 N mm^{-2})		550 days
Fatigue strength for 10^8 reversals at 15°C		± 2.5 MPa
Elastic modulus (tension)		
As cast		41.6 GPa
Self annealed		44.3 GPa
Impact strength		
0°C		44.1 J
150°C		22.7 J

Note: Mechanical properties are very dependent on the rate of loading.

Table 22.48 TIN-RICH SOLDERS—MECHANICAL PROPERTIES (BULK)

		Nominal composition (wt %)				
Properties	63Sn37Pb	60Sn40Pb	50Sn50Pb	95Sn5Sb	96.5Sn3.5Ag	62Sn36Pb2Ag
Tensile strength (N mm^{-2}) (Tested at 0.05 mm/min)						
at 20°C	na	19	na	31	37	43
at 100°C	na	4	na	20	na	19
Elongation (%)						
at 20°C	na	135	na	25	31	7
at 100°C	na	>100	na	21	na	na
Hardness (HB)	17	16	14	15	15	na
Young's modulus (GPa)	na	29.99	na	49.99	na	22.96
Creep strength (MPa) for life of 1000 h						
at 20°C	na	29	na	21	22	na
at 100°C	na	4.5	na	9	na	27
Impact strength (J)	20	20	21	27	na	na

Note: Tensile strength, hardness etc., are very dependent on the rate of loading and temperature of testing.
na = not available.

Table 22.49 PROPERTIES OF SOME TIN-RICH ALLOYS

Mechanical properties

Nominal composition (wt. %)							*Compressive yield* 0.125% set		*Strength* (MPa) 0.25% set					
Sn	Sb	Cu	Pb	Bi	*UTS* (MPa)	*Elongation* (%)	20°C	100°C	20°C	100°C	*Hardness* (HB)	*Liquidus* (°C)	*Solidus* (°C)	*Density* ($g\,cm^{-3}$)
Rem	4.5	4.5	—	—	64 Chillcast 62 Die cast	9 Chillcast 2 Die cast	30.3	17.9	88.2	47.6	17 Chillcast	371	223	7.34
Rem	7.5	3.5	—	—	77 Chillcast	18 Chillcast	42	21	103	60	24 Chillcast	354	241	7.39
Rem	8	8	—	—	69 Die cast	1 Die cast	46	21	117	6.8	27 Chillcast 30 Die cast	422	240	7.45
Rem	12	3	10	—	27 Chillcast	na	38.3	14.8	111.4	47.6	27 Chillcast	306	184	7.53
Rem	15	2	18	—	22.5 Chillcast	na	34	14	103	46	22.5 Chillcast	296	181	7.75
Rem	13	5	—	—	69 Die cast	1	na	na	na	na	29 Die cast	na	na	na
Rem	6	2	—	—	64.7 Chillcast 59 Annealed for 1 h air cooled 52 Cold rolled, 32% reduction	24 Chillcast 40 Annealed for 1 h air cooled 52 Cold rolled, 32% reduction	na	na	na	na	23.8 Chillcast 9.5 Annealed for 1 h air cooled 8 Cold rolled, 32% reduction	295	244	7.28
Rem	—	—	—	40	55	200 Slow loading	na	na	na	na	22	170	138	8.2

Note: Tensile strength, hardness, etc, are very dependent on the rate of loading and casting conditions.
Properties are typical values.
na = not available.

Next Page

FURTHER READING

A. General

1. 'Tin and its Alloys', E. S. Hedges, Edward Arnold, London, 1960.
2. 'Tin and Its Alloys and Compounds', B. T. K. Barry and C. J. Thwaites, Ellis Horwood Ltd, Chichester, England, 1983.
3. 'Metals Handbook', 10th Edition, Vol. 2, 'Properties and Selection, Non Ferrous Alloys and Pure Metals', American Society for Metals, Ohio, 1990.
4. Goodfellow, 'Materials for Research and Development', Catalogue 8, 1987/88.

B. Bearing Alloys

5. 'Babbit Alloys for Plain Bearings', International Tin Research Institute (ITRI), Publication No. 149.
6. 'Developments in Plain Bearing Technology', *Tribologia e Lubrificazione*, Vol. 10, p94, 1975, ITRI Publication No. 513.
7. 'Materials for Plain Bearings', G. C. Pratt, *International Metallurgical Review*, no. 174, Vol. 18, p62, 1973.
8. 'Tribology Handbook', M. J. Neale, Butterworth, London, 1973.
9. 'Tin-Base Casting Alloys', C. J. Evans, *Engineering Materials and Design*, p15, Aug, 1975.
10. 'Tin in Plain Bearings', ITRI Publication no. 595.

C. Pewter

11. 'Working with Pewter', ITRI Publication no. 566.
12. 'Modern Pewter', S. Charron, Van Nostrand Rheinhold, New York, 1973.
13. 'The Techniques of Pewtersmithing', C. Hull and J. Murrel, B.T. Batsford Ltd., London, 1984.

D. Fusible Alloys

14. 'Fusible Alloys Containing Tin', ITRI Publication no. 175.
15. 'The Properties of Tin Alloys', ITRI Publication no. 155.
16. 'Fusible Alloys for Industry', C. J. Evans, *Tin and its Uses*, no. 133, 1982, p10.
17. 'Fusible Alloys', *Engineering*, 1981, **221**(11), ppl.

E. Solder Alloys

18. 'Soft Soldering Handbook', ITRI Publication no. 533.
19. 'Solder Alloy Data', ITRI Publication no. 656.
20. 'Soldering Handbook', B. M. Allen, Iliffe, London, 1969.
21. 'Solders and Soldering', H. H. Manko, McGraw-Hill, New York, 1964.
22. 'Soldering Manual' American Welding Society, New York, 2nd Edition 1979.
23. 'Das Löten für den/Praktiker', R. Strauss, Franzis-Verlag, München, 1978.

22.10 Steels

Note for 8th edition: the tables in this section refer to standards that have been recently superceded by European (EN) and international (ISO) standards. The older British standards have been retained for information. Chapter 1 list equivalent designations in other systems for some alloys. For full information the relevant specifications must be consulted.